2026 edition

Deep Learning for Music Analysis and Generation

# Fundamentals for Musical Audio

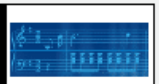
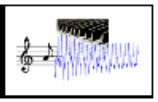
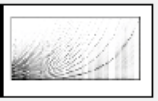
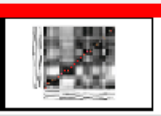
**Yi-Hsuan Yang** Ph.D.
yhyangtw@ntu.edu.tw

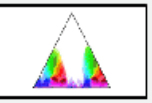
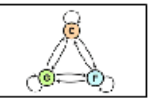
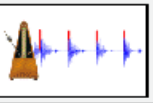
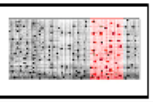
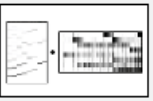
# Outline

- Time and frequency representations for music
- Math in STFT

# FMP Notebook

https://www.audiolabs-erlangen.de/resources/MIR/FMP/C1/C1.html
https://www.audiolabs-erlangen.de/resources/MIR/FMP/C2/C2.html

| Part | Title | Notions, Techniques & Algorithms | HTML | IPYNB |
|---|---|---|---|---|
| B  | Basics | Basic information on Python, Jupyter notebooks, Anaconda package management system, Python environments, visualizations, and other topics | [html] | [ipynb] |
| 0  | Overview | Overview of the notebooks (https://www.audiolabs-erlangen.de/FMP) | [html] | [ipynb] |
| 1  | Music Representations | Music notation, MIDI, audio signal, waveform, pitch, loudness, timbre | [html] | [ipynb] |
| 2  | Fourier Analysis of Signals | Discrete/analog signal, sinusoid, exponential, Fourier transform, Fourier representation, DFT, FFT, STFT | [html] | [ipynb] |
| 3  | Music Synchronization | Chroma feature, dynamic programming, dynamic time warping (DTW), alignment, user interface | [html] | [ipynb] |

| Part | Title | Notions, Techniques & Algorithms | HTML | IPYNB |
|---|---|---|---|---|
| 4  | Music Structure Analysis | Similarity matrix, repetition, thumbnail, homogeneity, novelty, evaluation, precision, recall, F-measure, visualization, scape plot | [html] | [ipynb] |
| 5  | Chord Recognition | Harmony, music theory, chords, scales, templates, hidden Markov model (HMM), evaluation | [html] | [ipynb] |
| 6  | Tempo and Beat Tracking | Onset, novelty, tempo, tempogram, beat, periodicity, Fourier analysis, autocorrelation | [html] | [ipynb] |
| 7  | Content-Based Audio Retrieval | Identification, fingerprint, indexing, inverted list, matching, version, cover song | [html] | [ipynb] |
| 8  | Musically Informed Audio Decomposition | Harmonic/percussive separation, signal reconstruction, instantaneous frequency, fundamental frequency (F0), trajectory, nonnegative matrix factorization (NMF) | [html] | [ipynb] |

# Outline

- **Time and frequency representations for music**
- Math in STFT

# Audio Waveforms

https://www.audiolabs-erlangen.de/resources/MIR/FMP/C1/C1S3_Waveform.html

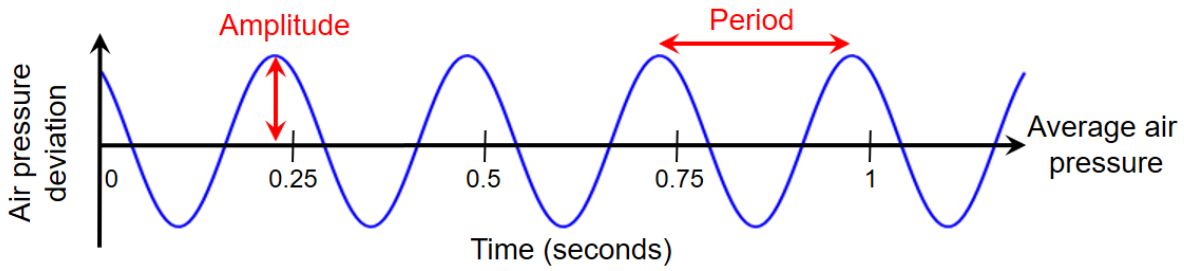
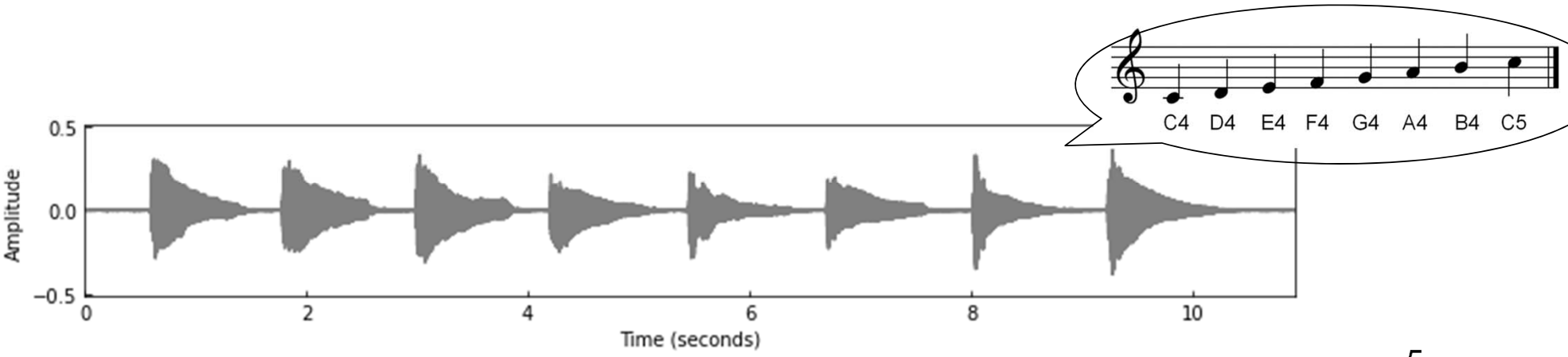



Figure 1.19 from [Müller, FMP, Springer 2015]



# What Do We Perceive in Musical Audio?

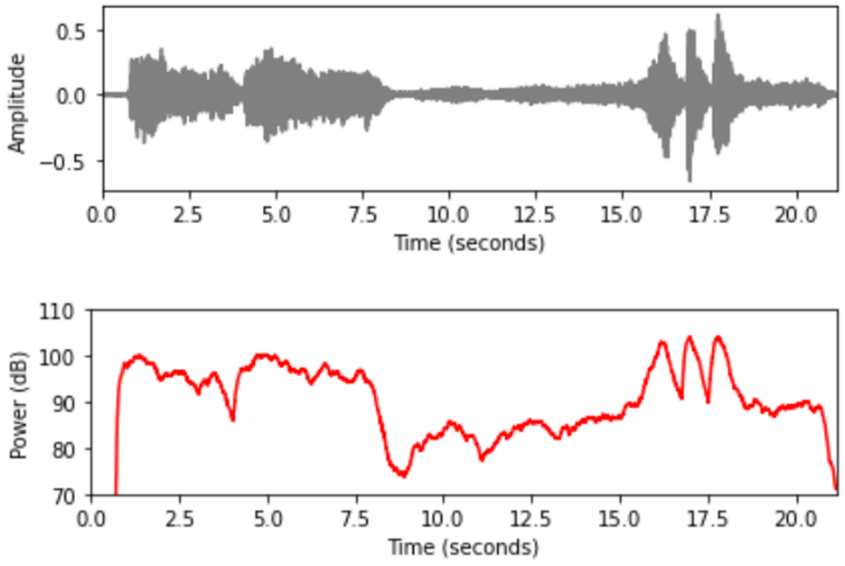
- Loudness
- Spatial information
- Timbre / instrument / singer identity
- Pitch / melody
- Chord / harmony
- Tempo / rhythm
- Style
- Emotion

- Others (e.g., “The power of music compounds over time through repeated listening. Culture builds around shared experiences.”)

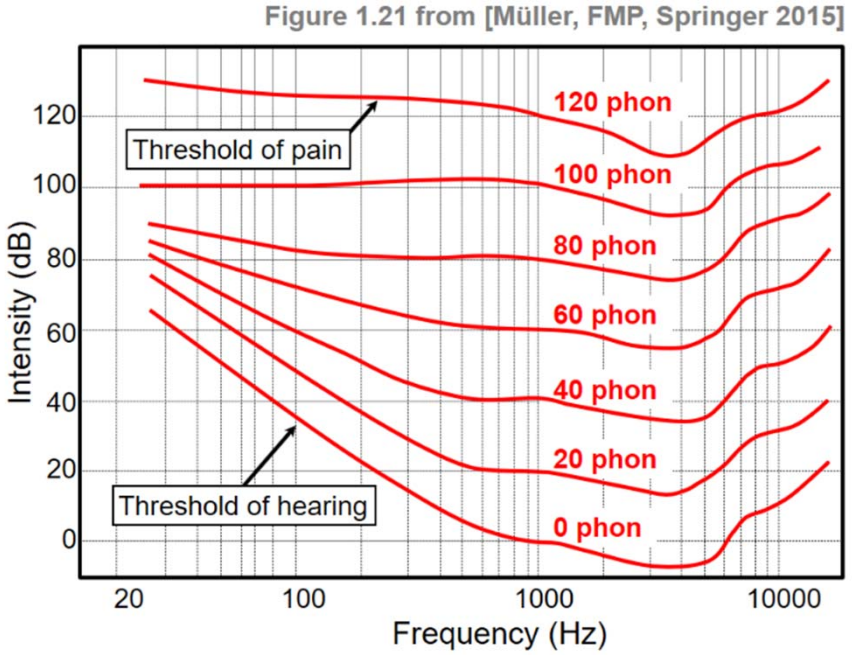
# Dynamics, Intensity, and Loudness

https://www.audiolabs-erlangen.de/resources/MIR/FMP/C1/C1S3_Dynamics.html

- **Intensity**: a physical property
  - Other names: power (dB); energy
  - **Peak** vs. root mean square (**RMS**)
  - Dynamic range: dB max/min difference



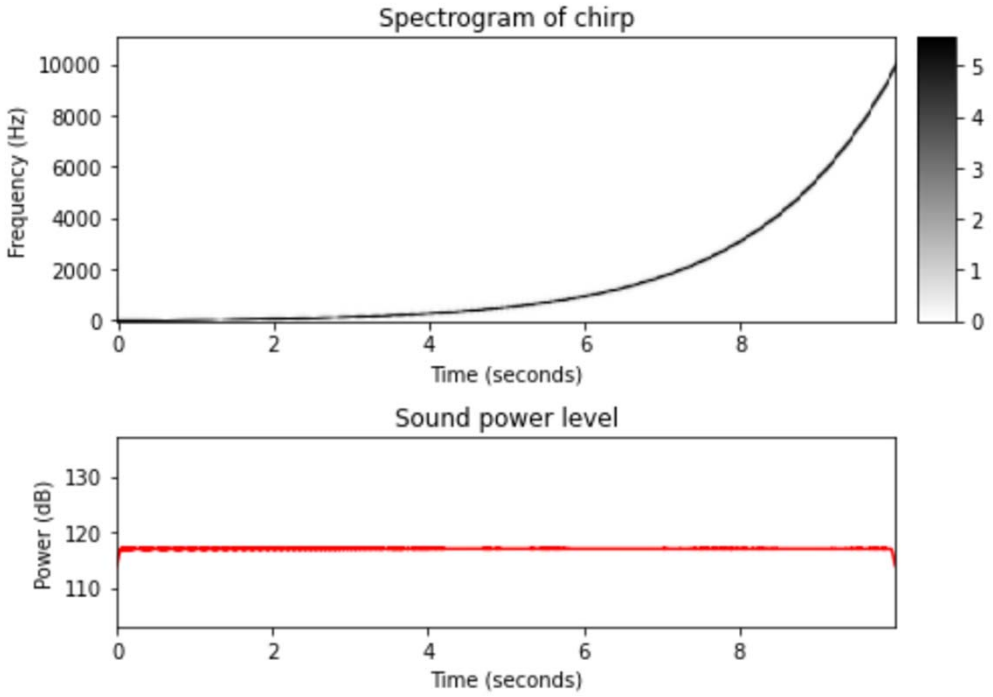

- **Loudness**: a perceptual property
  - **LUFS**: frequency-weighted average
  - volume: user control of loudness



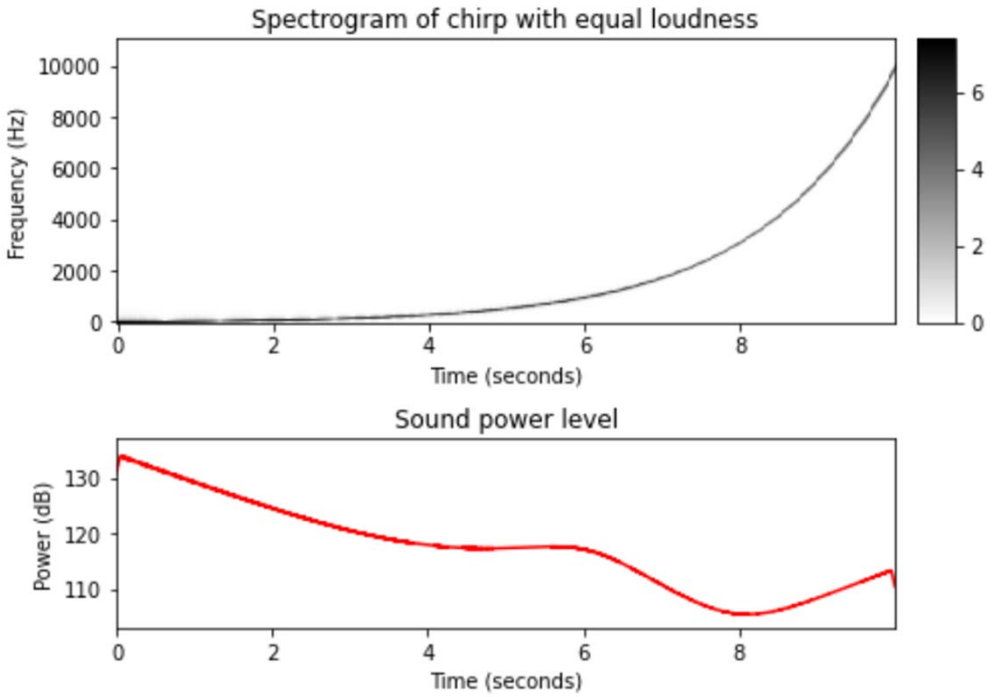
# Dynamics, Intensity, and Loudness

https://www.audiolabs-erlangen.de/resources/MIR/FMP/C1/C1S3_Dynamics.html

- Chirp signal with equal power




- Chirp signal with equal loudness



# Dynamics, Intensity, and Loudness

https://www.audiolabs-erlangen.de/resources/MIR/FMP/C1/C1S3_Dynamics.html

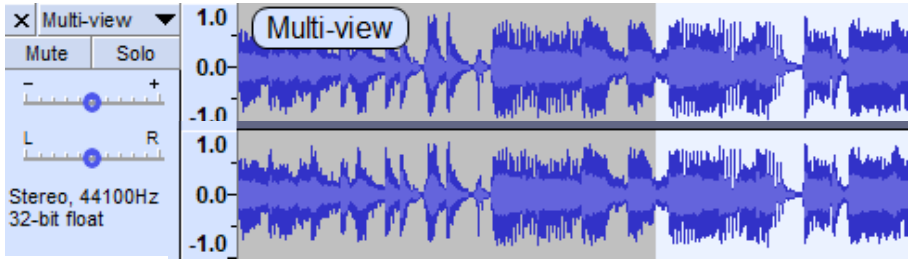
- Mixing, mastering


- Audio normalization
  - E.g., https://github.com/csteinmetz1/pyloudnorm

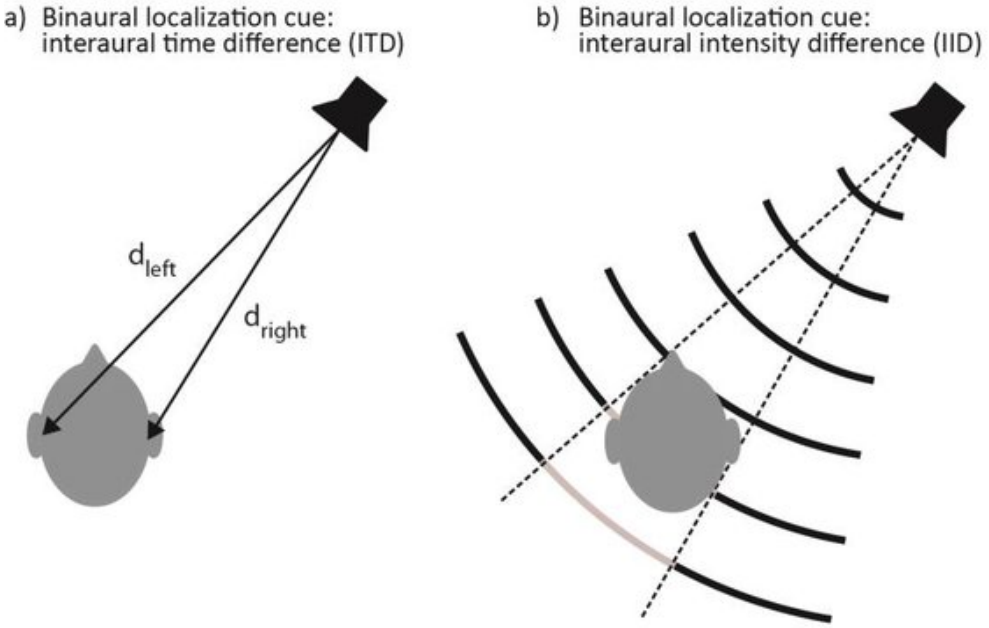
```
import soundfile as sf
import pyloudnorm as pyln

data, rate = sf.read("test.wav") # load audio

# peak normalize audio to -1 dB
peak_normalized_audio = pyln.normalize.peak(data, -1.0)

# measure the loudness first
meter = pyln.Meter(rate) # create BS.1770 meter
loudness = meter.integrated_loudness(data)

# loudness normalize audio to -12 dB LUFS
loudness_normalized_audio = pyln.normalize.loudness(data, loudness, -12.0)
```

# Spatial Information: Mono vs Stereo

- Monaural vs. binaural




- Our auditory system constructs a **3D** mental model of a soundscape
- **Localization** of sound objects
  - Azimuth, elevation, distance
- Perception of **room acoustics**




Source: https://www.researchgate.net/figure/Binaural-and-monaural-cues-used-for-sound-localization-a-For-sound-sources-off-the_fig1_299281975

# torchaudio

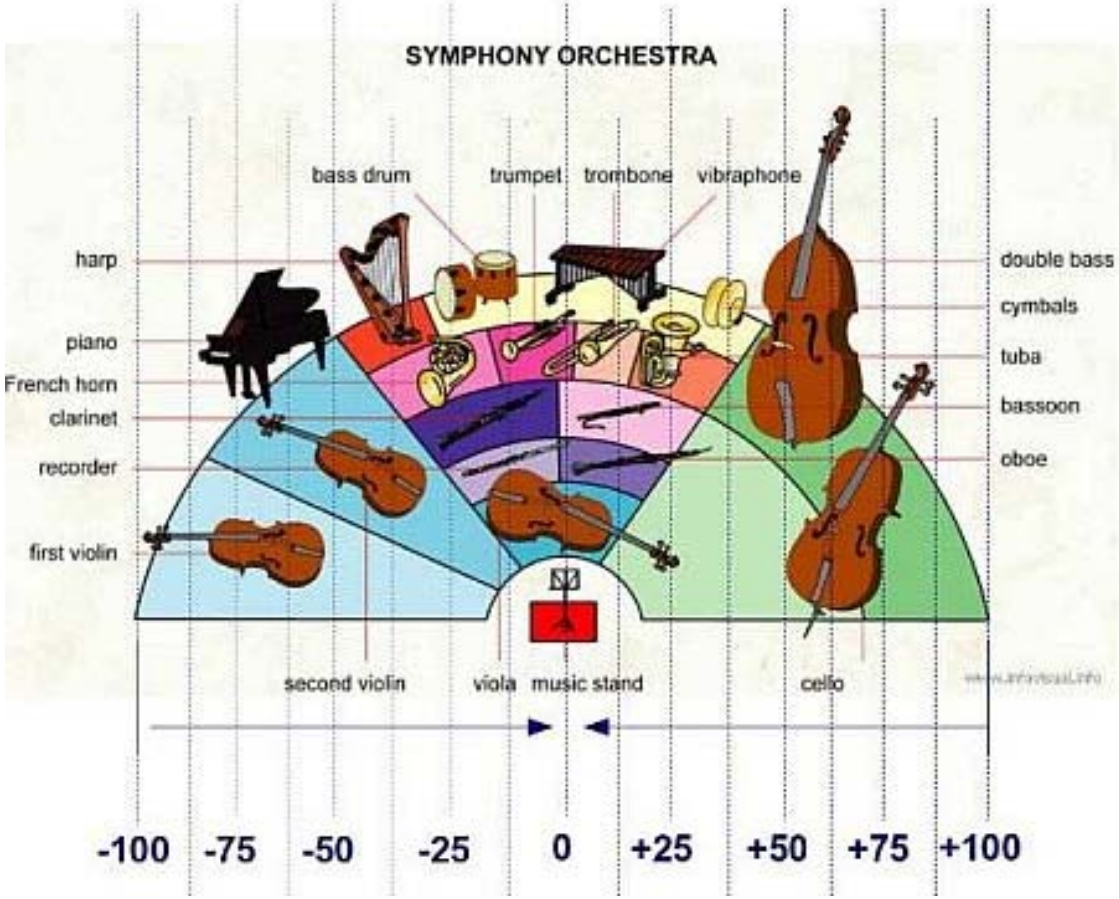
https://docs.pytorch.org/audio/stable/generated/torchaudio.load.html

https://docs.pytorch.org/audio/2.0.1/tutorials/audio_io_tutorial.html

```
waveform, sample_rate = torchaudio.load(SAMPLE_WAV)


def plot_waveform(waveform, sample_rate):
    waveform = waveform.numpy()
    num_channels, num_frames = waveform.shape
```

# Spatial Information


https://www.renegadeproducer.com/audio-panning.html




https://www.audiorecording.me/symphony-orchestra-panning-and-reverb-settings.html

# Timbre: Different Instruments

https://www.audiolabs-erlangen.de/resources/MIR/FMP/C1/C1S3_Timbre.html

- **Timbre** (tone color; tone quality)
  - Allow us to distinguish the musical tone (i.e., a single musical note) of different instruments, even if the tone is played at the same **pitch** and with the same **loudness**
  - Subjective & it's hard to describe
  - May be described with words such as *bright*, *dark*, *warm*, and *harsh*

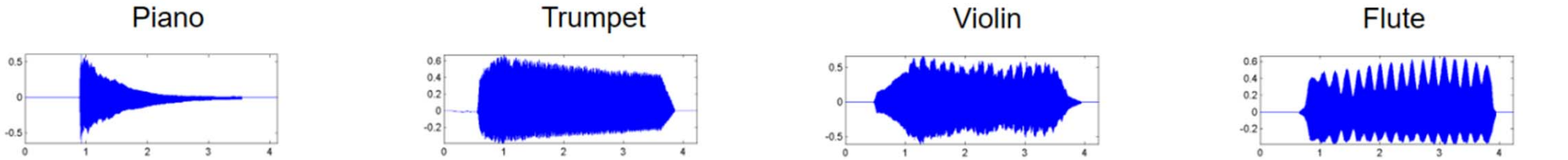



Figure 1.23 from [Müller, FMP, Springer 2015]

0:04 / 0:04 0:00 / 0:04 0:00 / 0:04 0:00 / 0:04

# Timbre: Temporal Aspect

- **ADSR**: attack (A), decay (D), sustain (S), and release (R)
  - Hard to analyze when there are concurrent notes

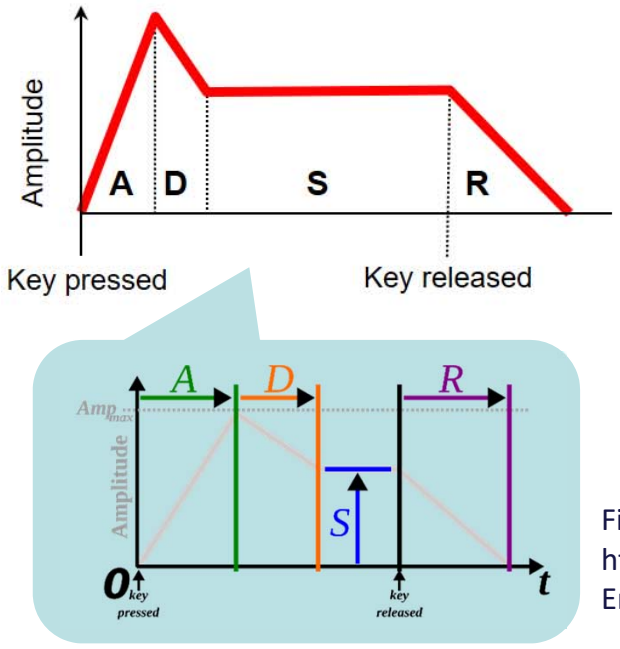
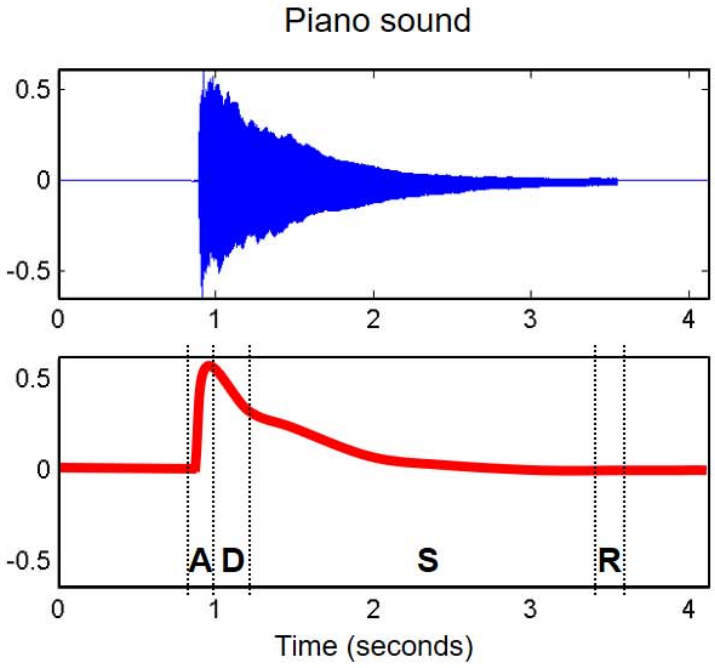



Figure from:
https://en.wikipedia.org/wiki/Envelope_(music)



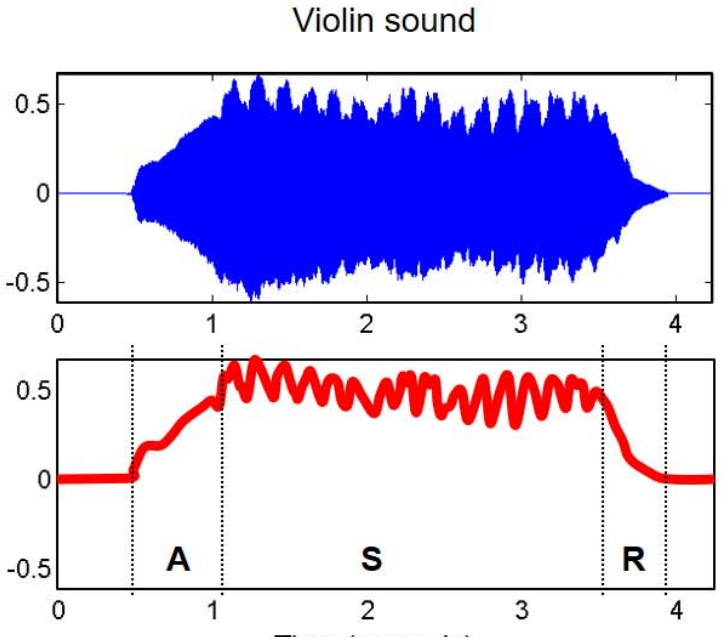
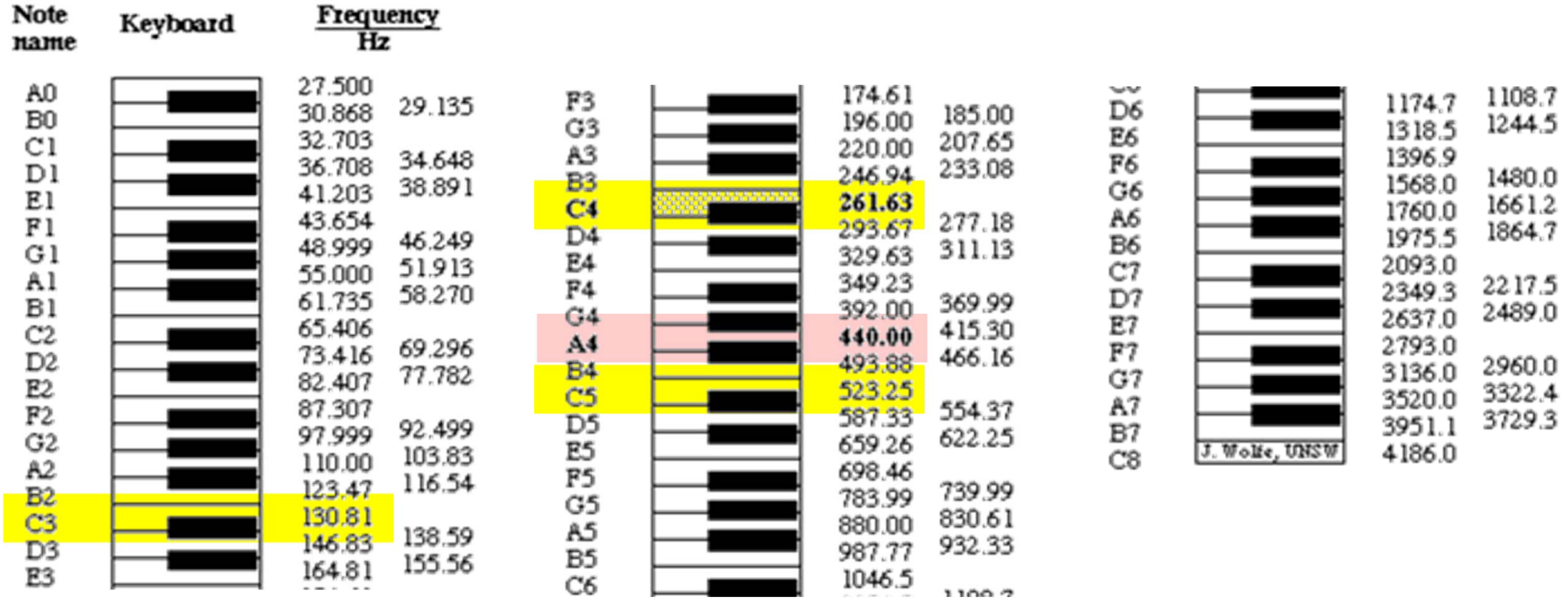




Figure 1.22b and Figure 1.23
from [Müller, FMP, Springer 2015]

# Notes and Pitches

(Assuming A4=440 Hz; but there are *other tunings*)

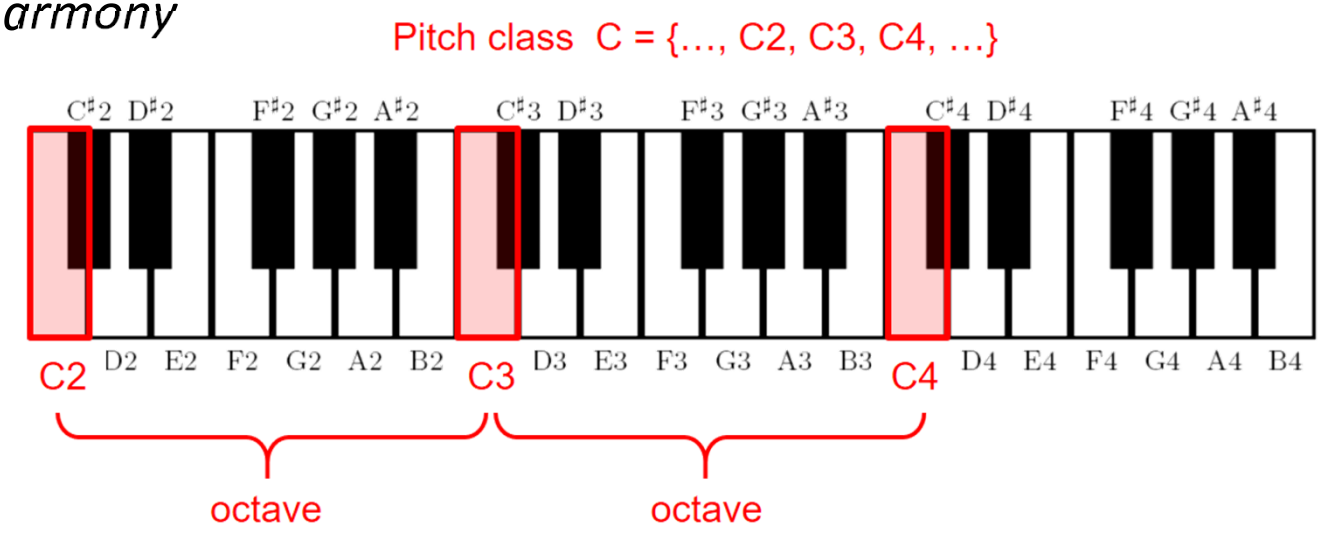
https://newt.phys.unsw.edu.au/jw/notes.html



# Notes and Pitches

https://www.audiolabs-erlangen.de/resources/MIR/FMP/C1/C1S1_MusicalNotesPitches.html

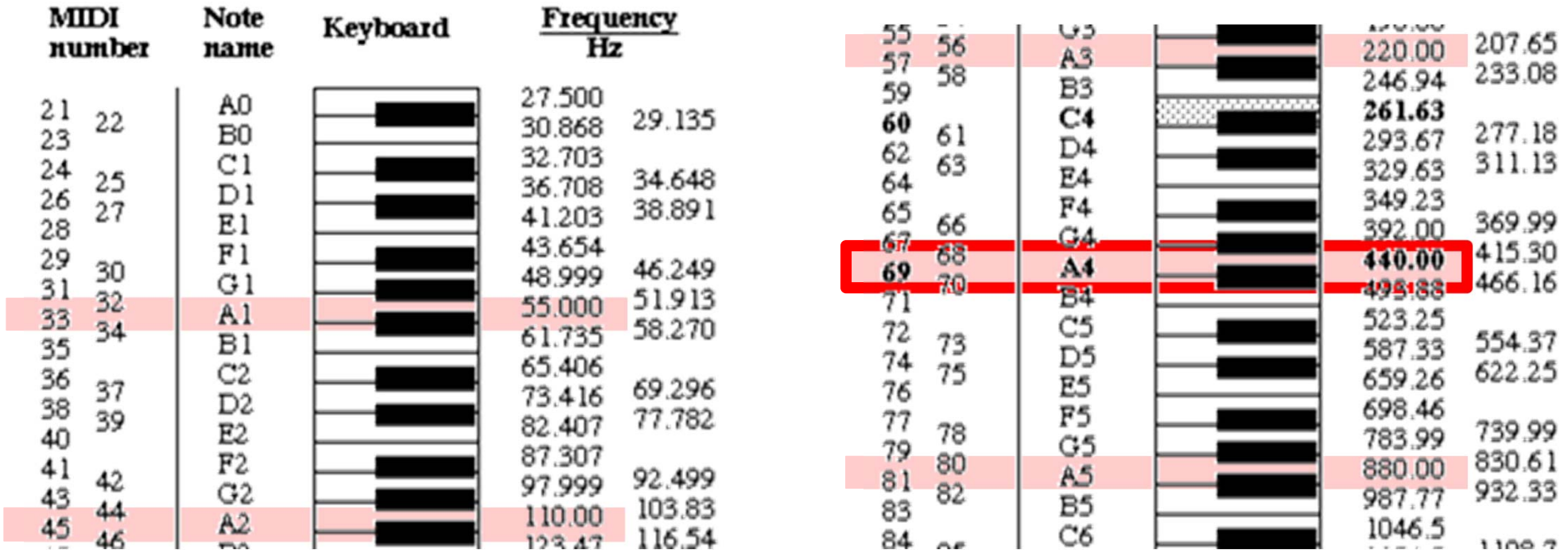
- **Pitch Class**
  - Two notes with *fundamental frequencies* in a ratio equal to *any power of* ***two*** (e.g., half, twice, or four times) are perceived as very **similar**
  - All notes with this kind of relation can be grouped under the same *pitch class*
  - Related to *chords/harmony*



# MIDI Note Numbers

https://en.wikipedia.org/wiki/MIDI_tuning_standard

$$f = 2^{(d-69)/12} \cdot 440\ \text{Hz} \qquad d = 69 + 12\log_2\left(\frac{f}{440\ \text{Hz}}\right)$$

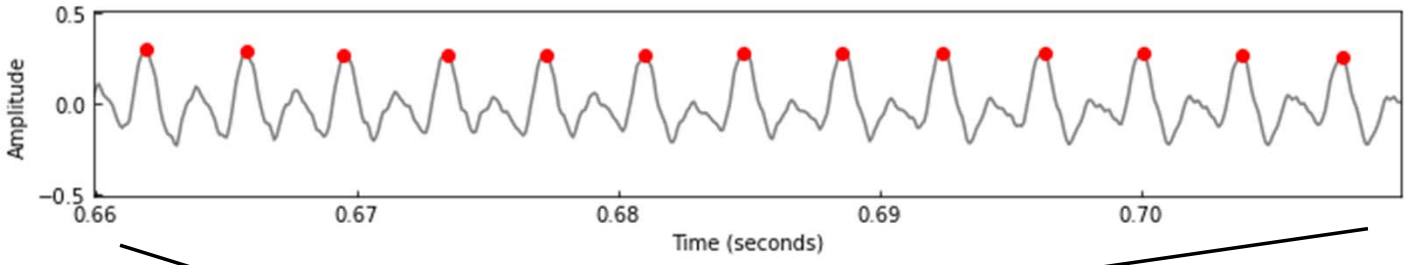


# Audio Waveforms (Cont')

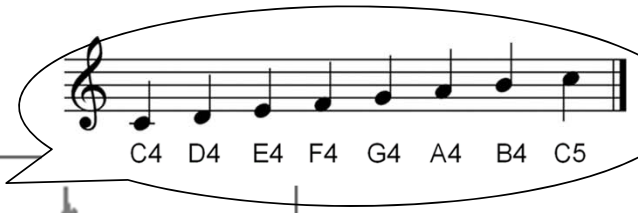
https://www.audiolabs-erlangen.de/resources/MIR/FMP/C1/C1S3_Waveform.html

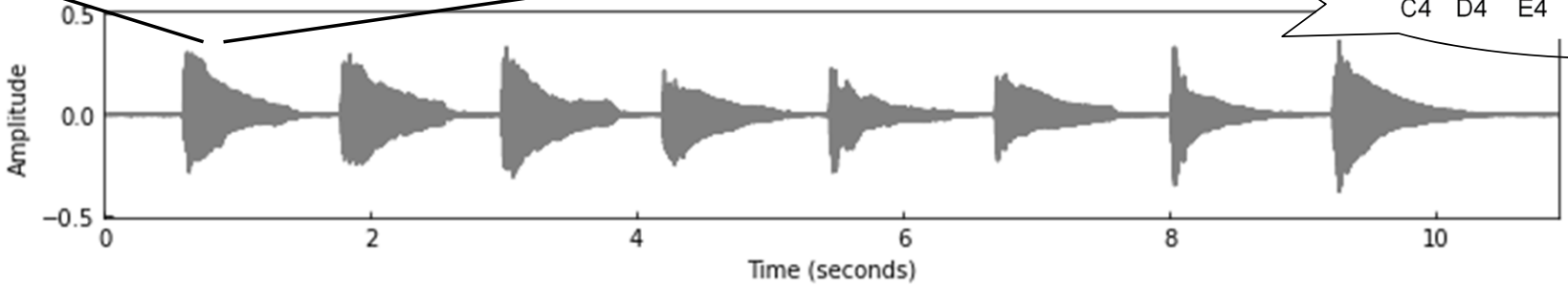
Zoom-in of the section between 0.66 and 0.71 seconds

- Highly repetitive
- 13 high-pressure (red) points → 20*13=260 Hz



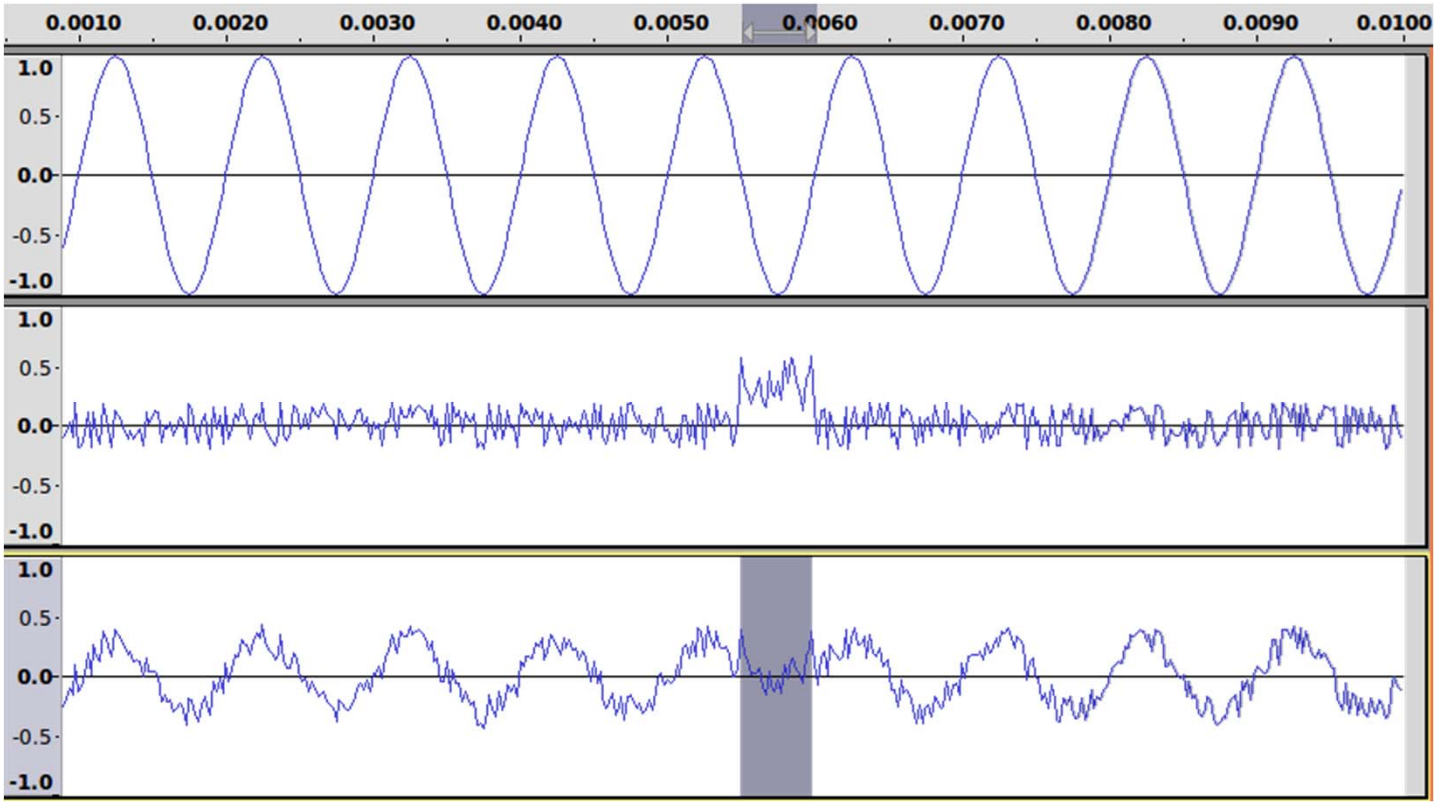




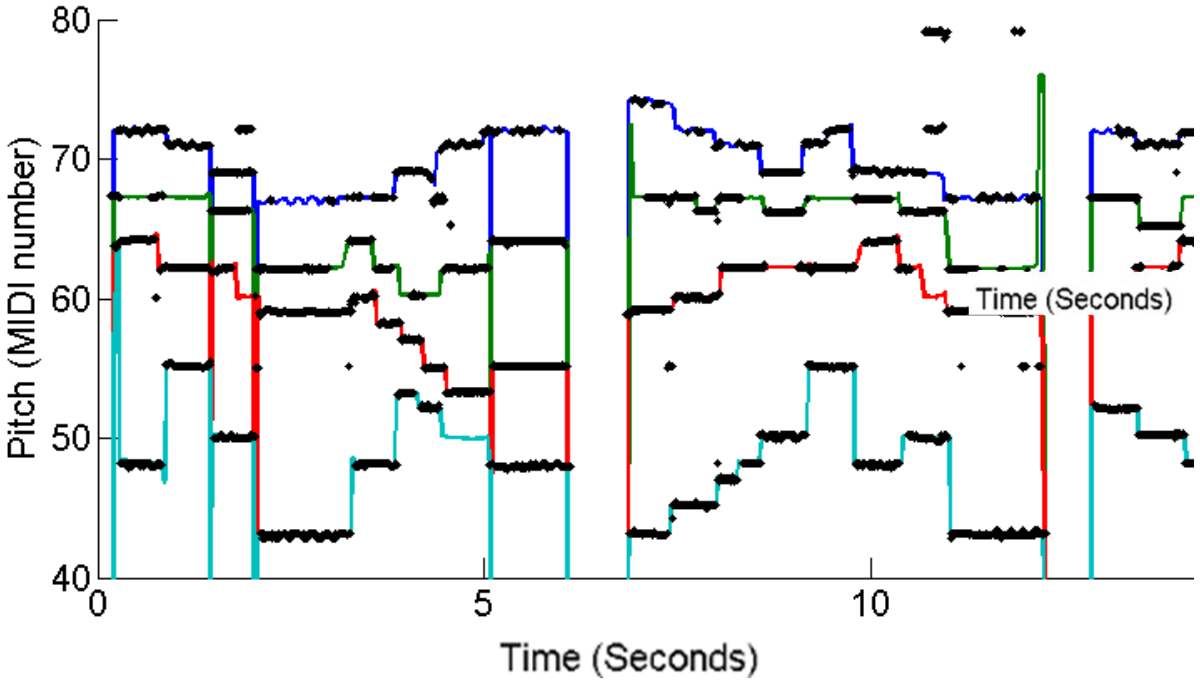
# Temporal Characteristics : Pitched Sound or Not

- **Zero-crossing rate**
  - How often a signal changes its sign
  - **Nosie**-like or not
  - Can be used in detecting the presence of vocals in speech signals
  - Cannot be used for estimating pitch due to noises and concurrent notes

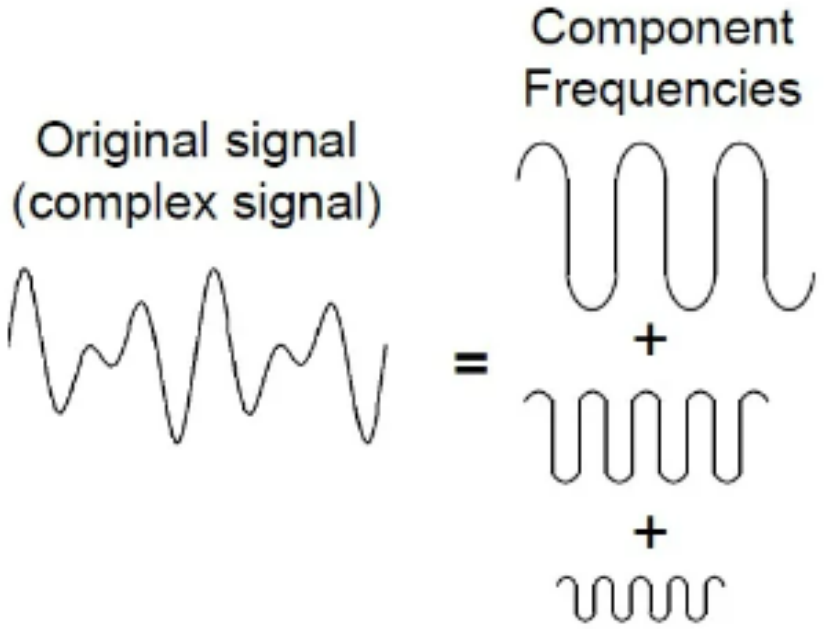



https://sandilands.info/sgordon/impact-of-noise-on-sine-wave-compared-to-square-wave

# Multiple Concurrent Pitches



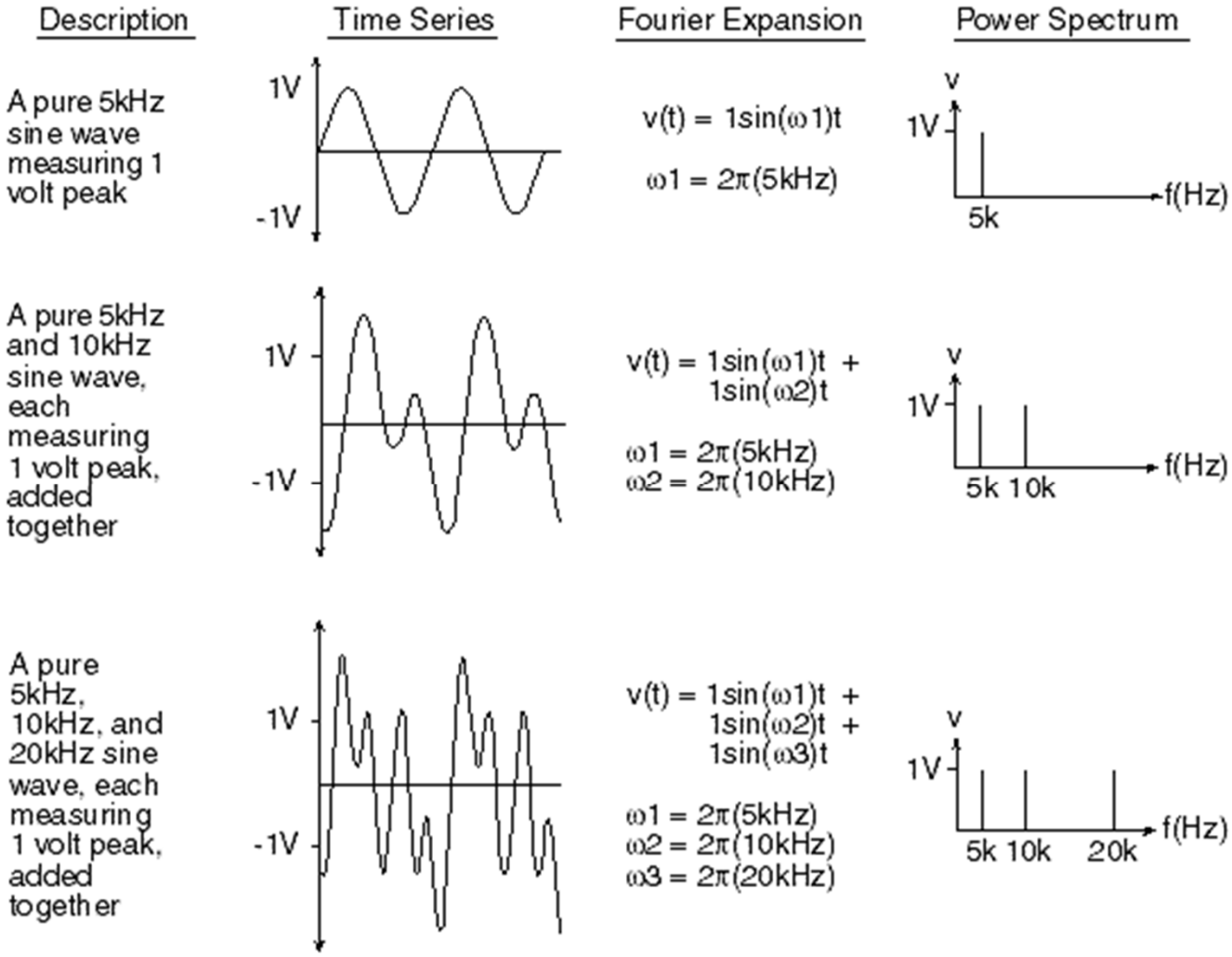

https://labsites.rochester.edu/air/projects/multipitch/multipitch.html




https://www.allaboutcircuits.com/technical-articles/an-introduction-to-filters/

# Frequency-domain Analysis

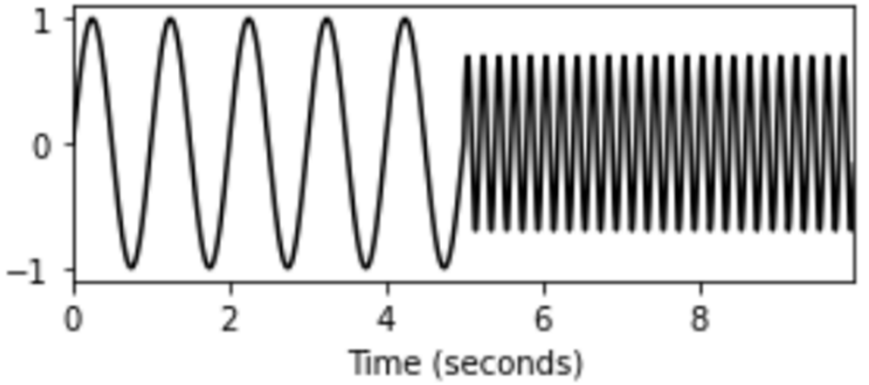
- Main subject of "Signals and Systems"
- **Fourier Transform**

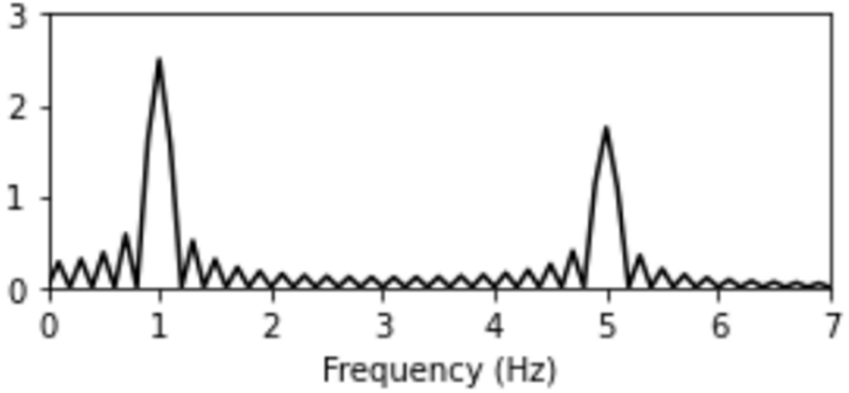


# Frequency-domain Analysis

https://www.audiolabs-erlangen.de/resources/MIR/FMP/C2/C2_STFT-Basic.html

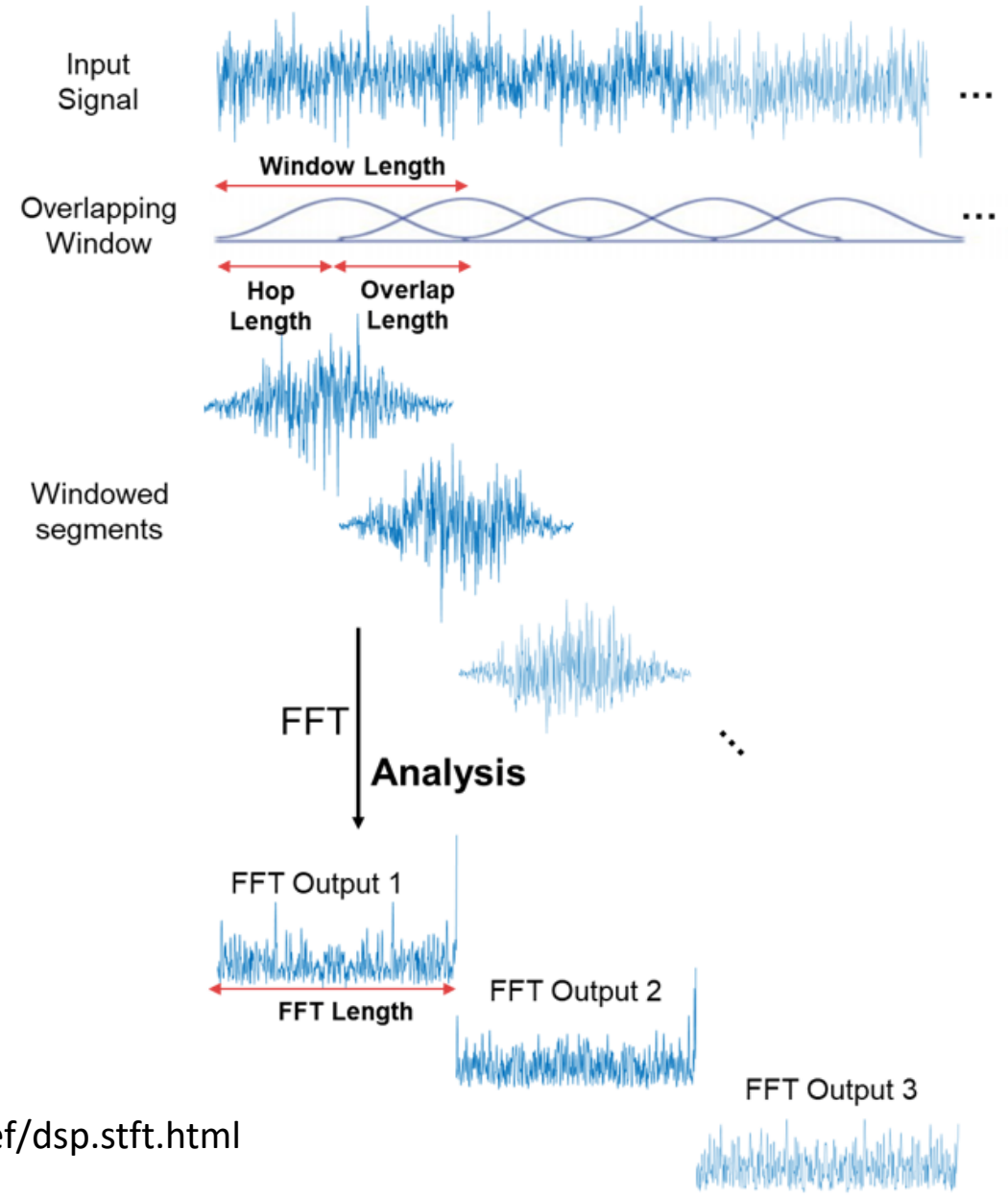
- However, Fourier Transform cannot localize temporal events



# Frequency-domain Analysis

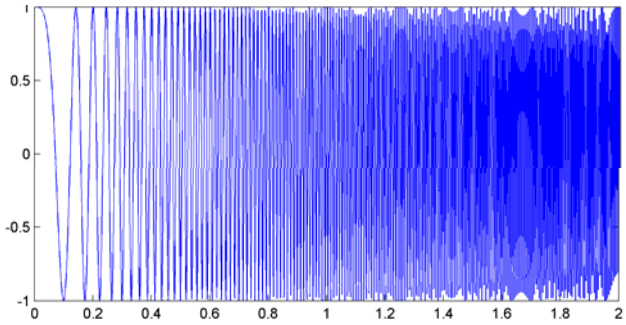
- Fourier Transform (FFT) cannot localize temporal events
- **Short-Time Fourier Transform** (STFT)
  - *Time-frequency representation* of signals
  - Slide **short windows** over the input signal and then compute the FFT locally for each windowed signal
- "**Window**": a short chunk of time




https://ww2.mathworks.cn/help/dsp/ref/dsp.stft.html

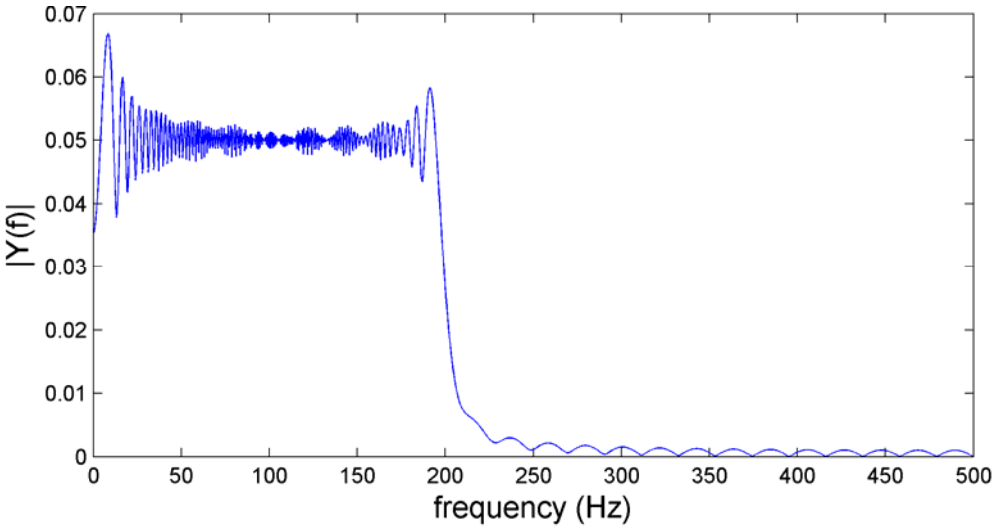
# FFT vs STFT

Time-domain **waveform**

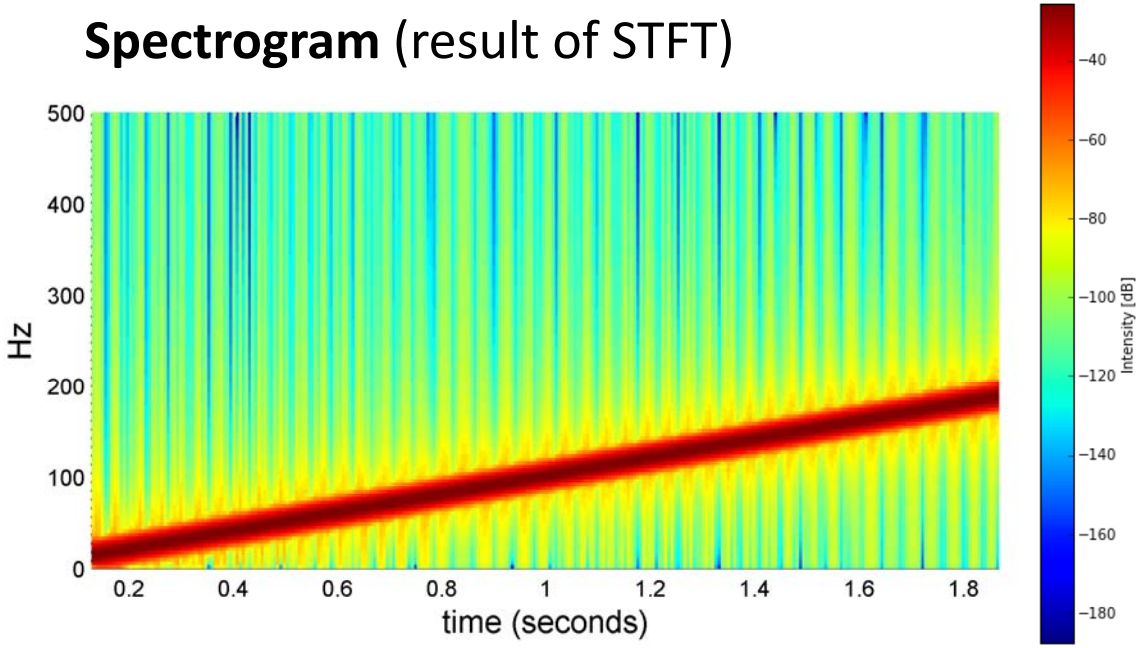



- **Waveform**: amplitude vs time
- **Spectrum**: amplitude vs frequency
- **Spectrogram**: time vs frequency; use *color* for amplitude

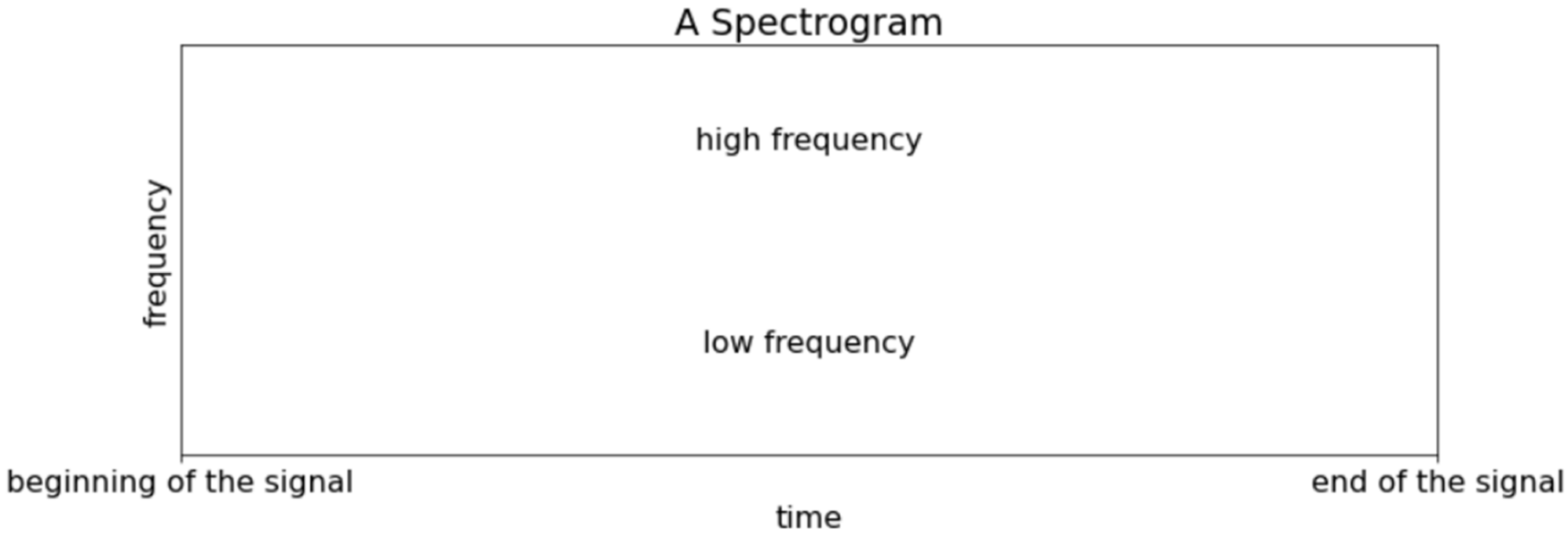
**Spectrum** (result of FFT)




**Spectrogram** (result of STFT)



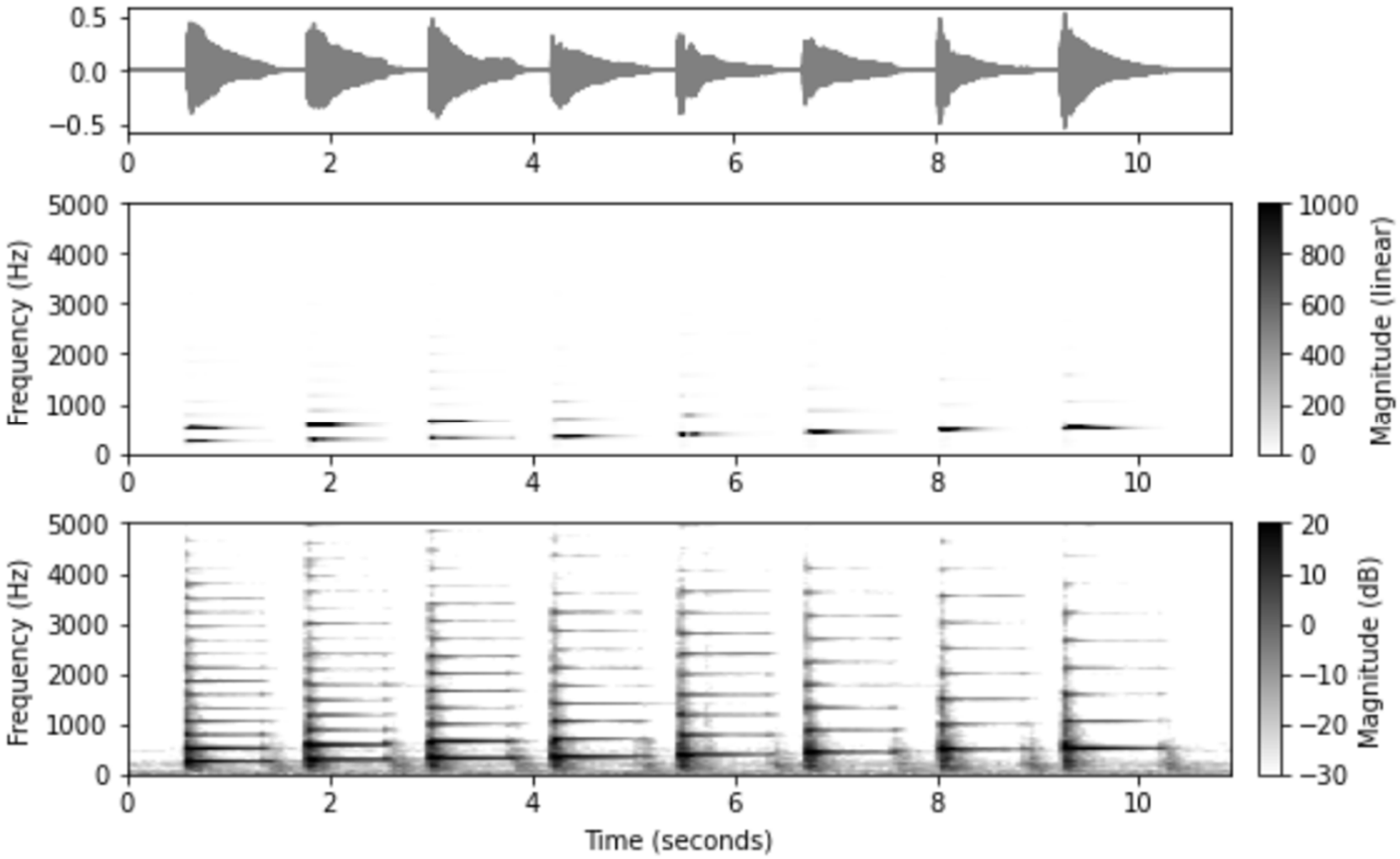
# Time-Frequency Analysis via STFT




https://music-classification.github.io/tutorial/part2_basics/input-representations.html

# Time-Frequency Analysis via STFT

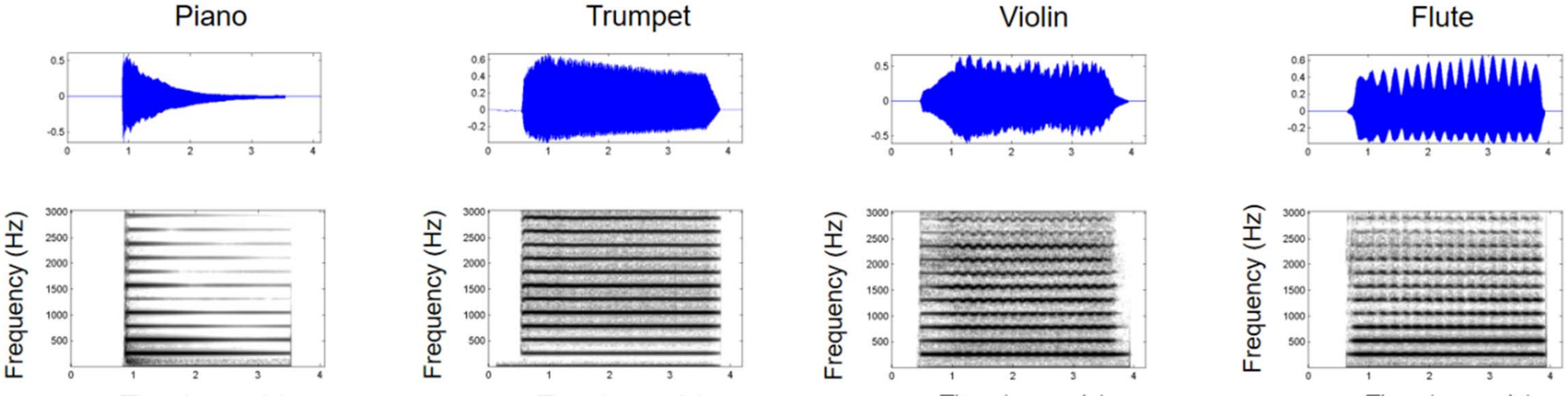
https://www.audiolabs-erlangen.de/resources/MIR/FMP/C2/C2_STFT-Basic.html




- **Time-frequency representation** of audio
  - Linear frequency, **linear** magnitude
  - Linear frequency, **logarithmic** magnitude (dB)

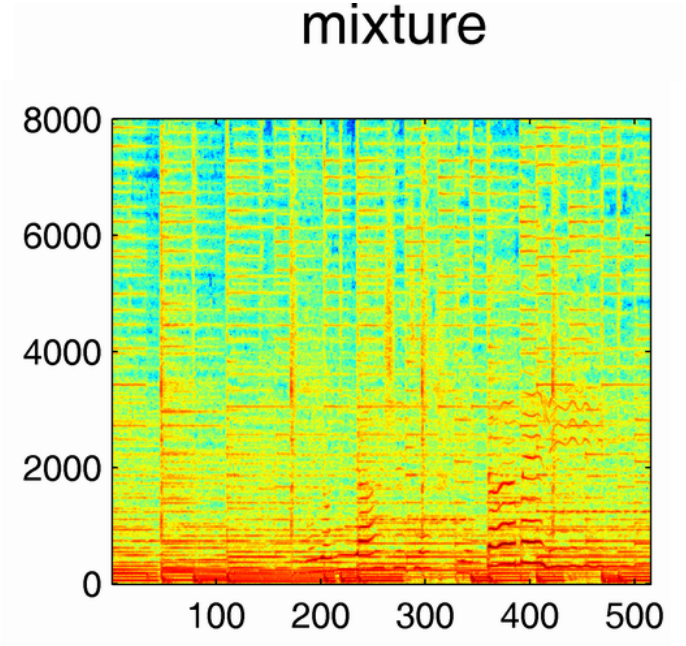
# Timbre: Spectral Aspect

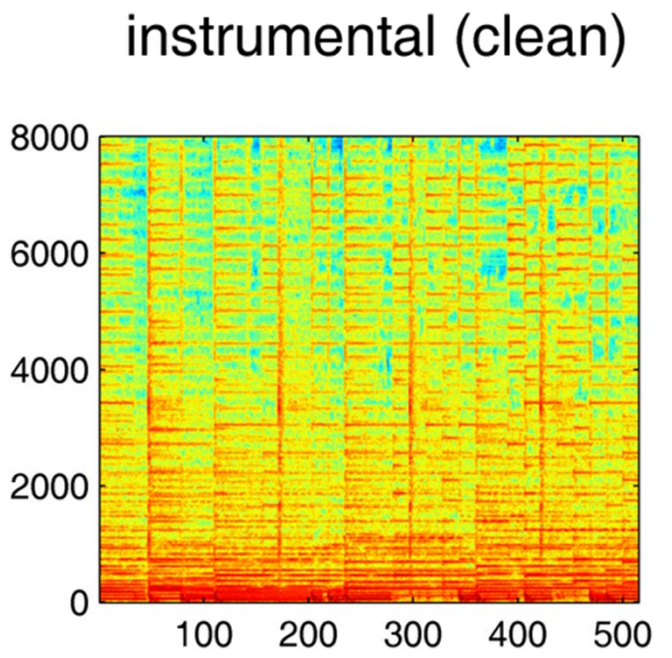
- Fundamental frequency (**F0**)
- **Partials** (harmonics): usually at frequencies which are **integer multiples** of F0
  - The partials in different instruments may exhibit **relative strengths**
  - The frequency may *deviate* from the ideal harmonics (this is called **inharmonicity**)

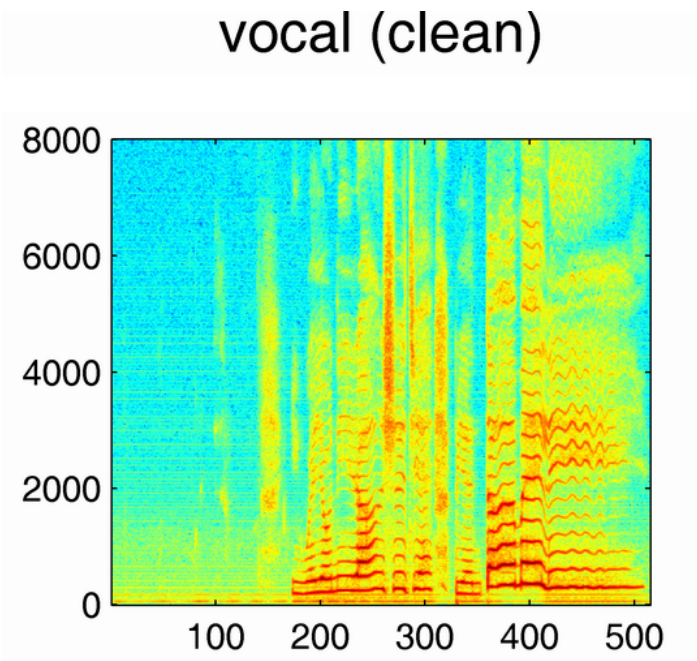


# Time-Frequency Analysis via STFT

- For tasks such as classification/detection and source separation

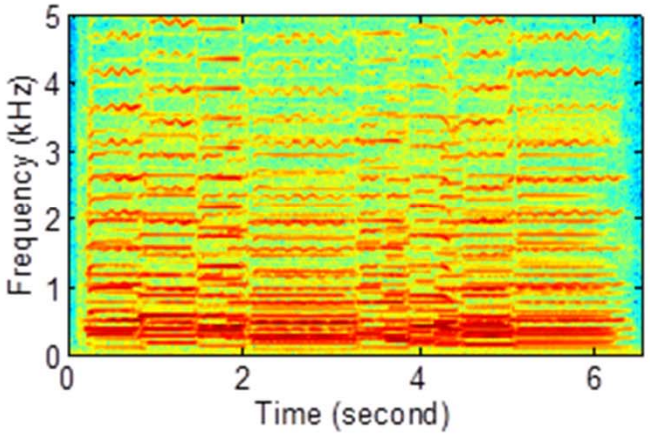


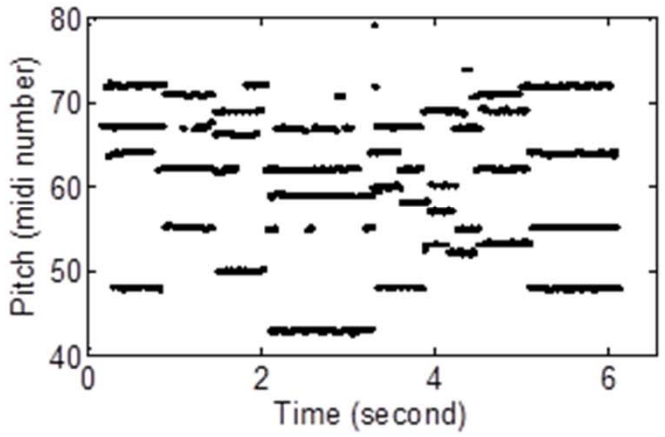



# Time-Frequency Analysis via STFT

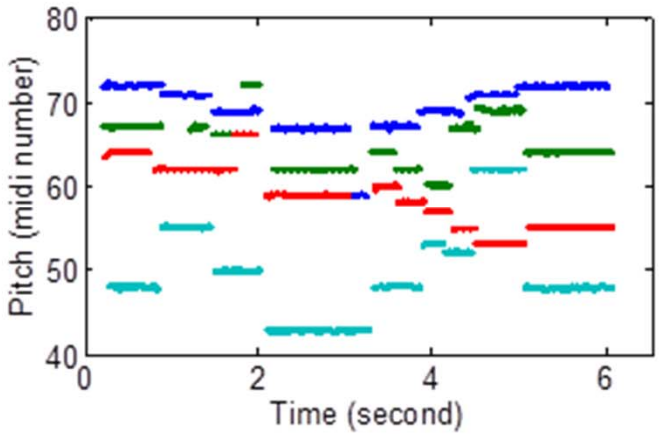
- For tasks such as multi-pitch estimation (MPE) and transcription




Spectrogram




Multi-pitch estimation results




MPE + instrument detection

https://labsites.rochester.edu/air/projects/multipitch/multipitch.html

# torchaudio. transforms.Spectrogram

https://docs.pytorch.org/audio/main/generated/torchaudio.transforms.Spectrogram.html

```
waveform, sample_rate = torchaudio.load("test.wav", normalize=True)

transform = torchaudio.transforms.Spectrogram(n_fft=800)

spectrogram = transform(waveform)
```

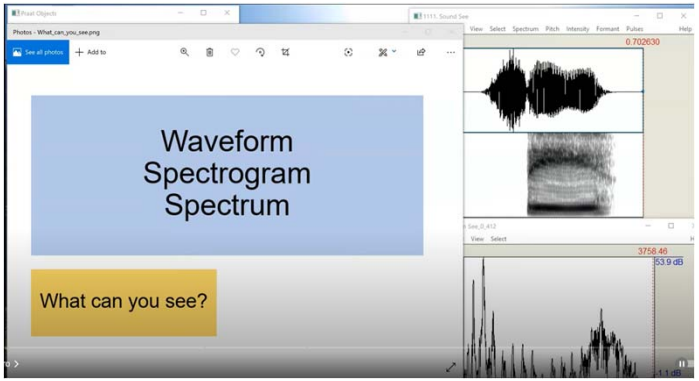
**Parameters**

- **n_fft** (*int, optional*) – Size of FFT, creates `n_fft // 2 + 1` bins. (Default: `400`)
- **win_length** (*int or None, optional*) – Window size. (Default: `n_fft`)
- **hop_length** (*int or None, optional*) – Length of hop between STFT windows. (Default: `win_length // 2`)

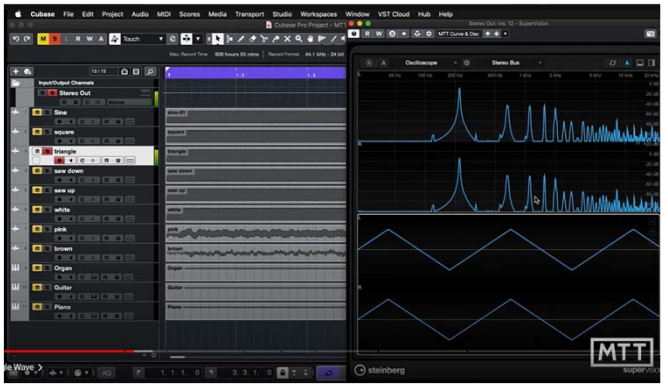
# Source of Copyleft Audio Recordings

- https://www.freesound.org/
  - https://www.freesound.org/people/acclivity/sounds/22347/
  - https://www.freesound.org/people/Rudmer_Rotteveel/sounds/316915/
  - https://www.freesound.org/people/Jaylew1987/sounds/321112/
  - https://www.freesound.org/people/mickel11/sounds/90803/

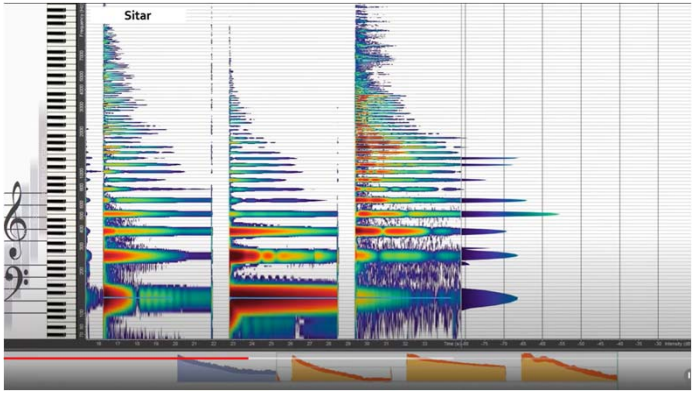
# Related Videos



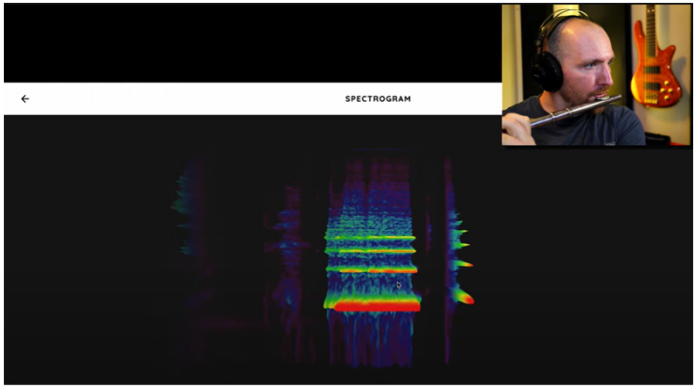

https://www.youtube.com/watch?v=2Hj1kAWVjLo




https://www.youtube.com/watch?v=fZ18C5RXDcc



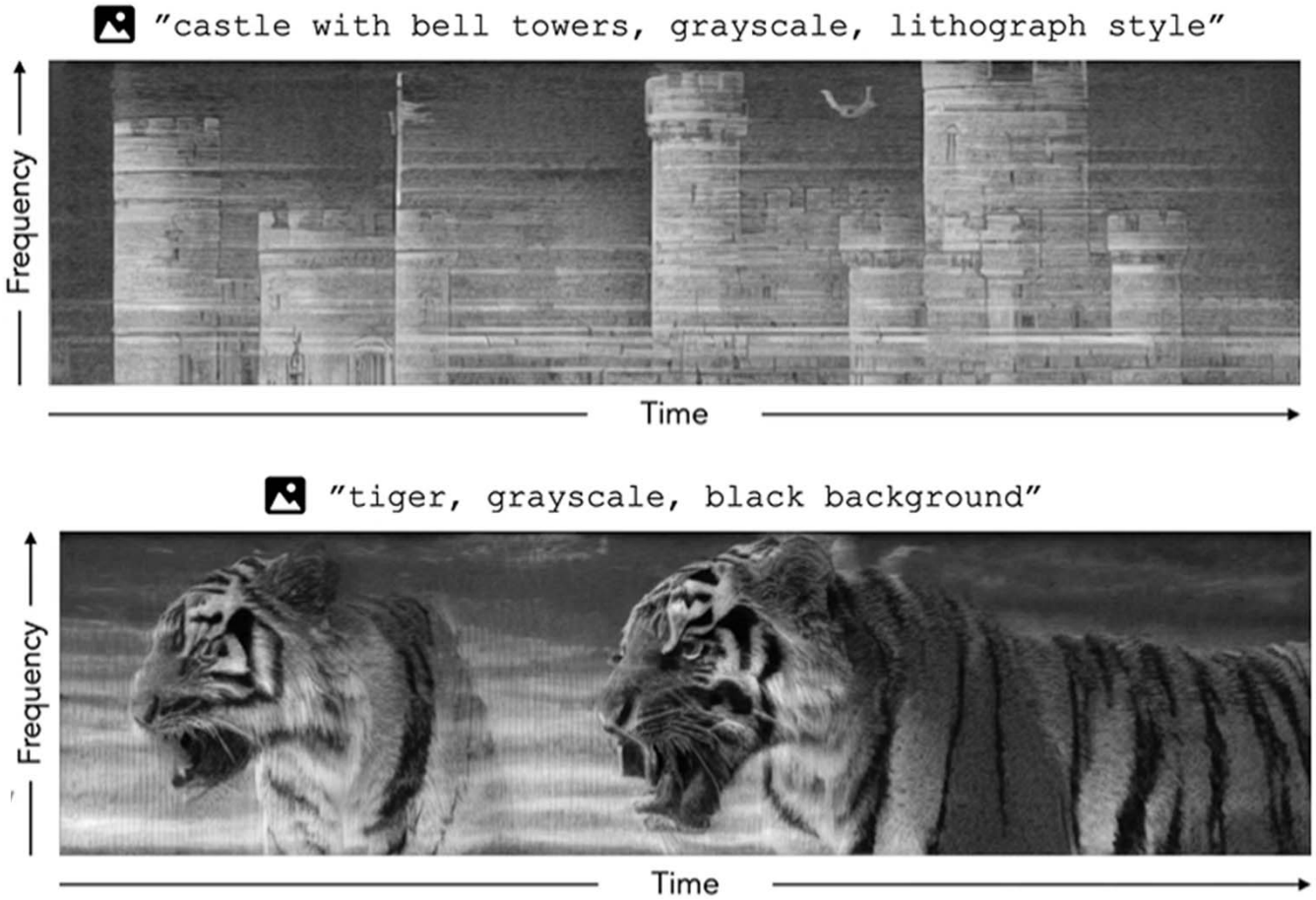

https://www.youtube.com/watch?v=VRAXK4QKJ1Q

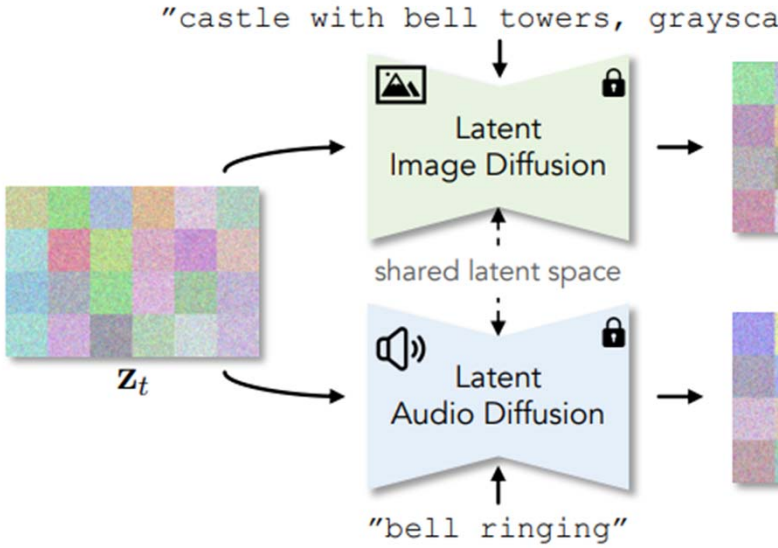



https://www.youtube.com/watch?v=jEO0GHU3xsU

# Fun Stuff: “Images that Sound”

https://ificl.github.io/images-that-sound/



# What Do We Perceive in Musical Audio?

- Loudness
- Spatial information
- Timbre / instrument / singer identity
- Pitch / melody
- Chord / harmony
- Tempo / rhythm
- Style
- Emotion

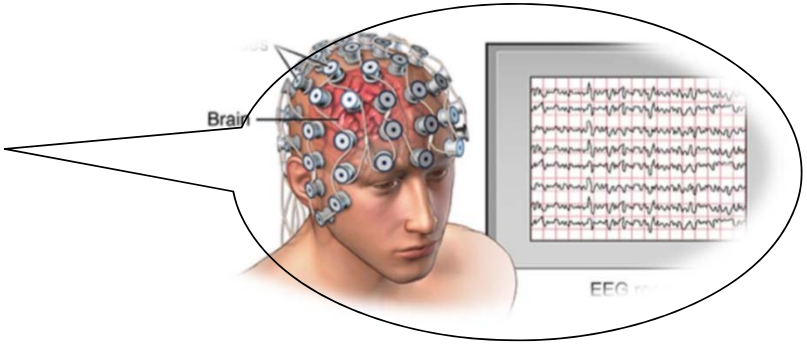
- Others (e.g., “The power of music compounds over time through repeated listening. Culture builds around shared experiences.”)

# Outline

- Time and frequency representations for music
- **Math in STFT**

# Sampling Rate

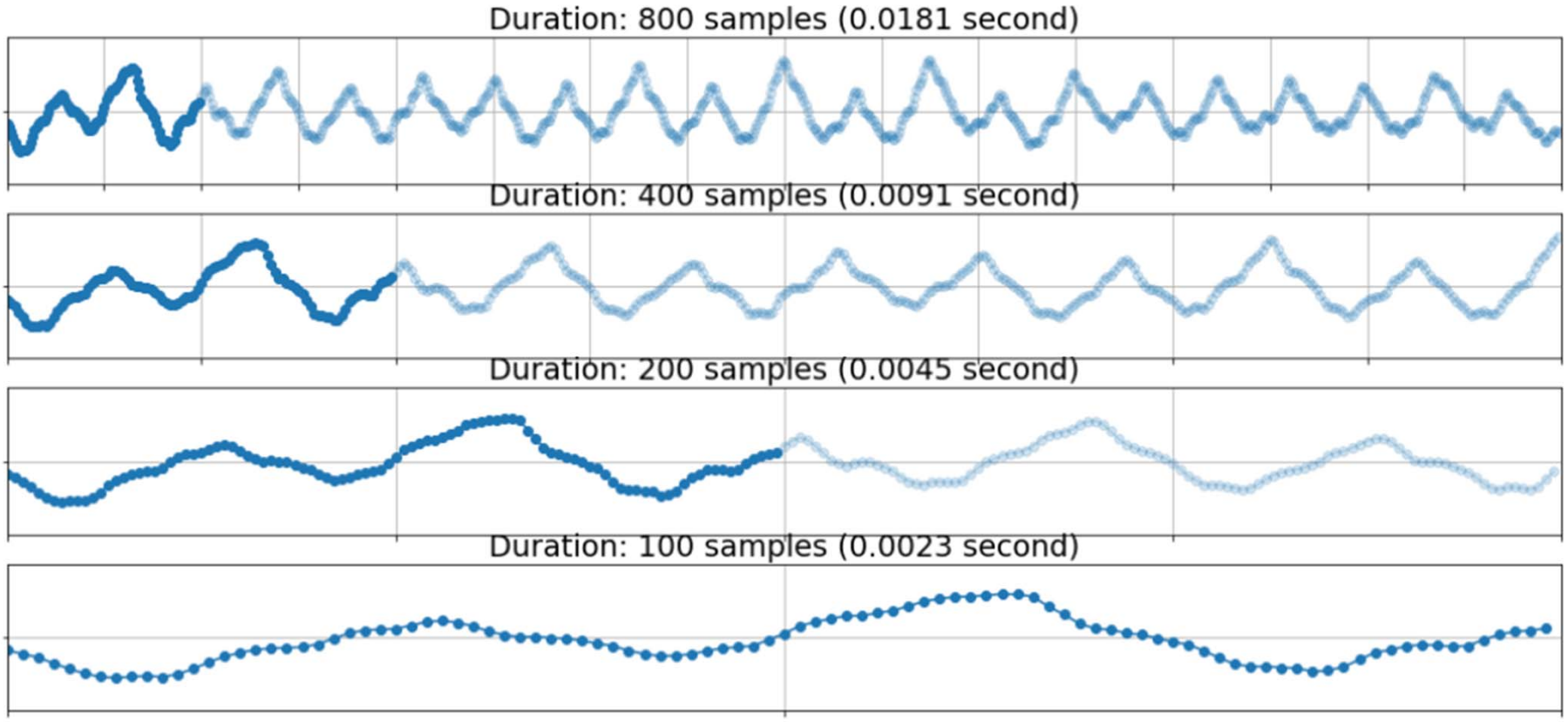
- Definition: number of samples per second
- Why: analog to digital
- Examples
  - EEG signal: **128** Hz
  - Telephone audio: **8k** Hz
  - Music audio: **44k** HZ




https://www.brightbraincentre.co.uk/electroencephalogram-eeg-brainwaves/

```
import soundfile as sf
import pyloudnorm as pyln

data, rate = sf.read("test.wav") # load audio
```

# Sampling Rate



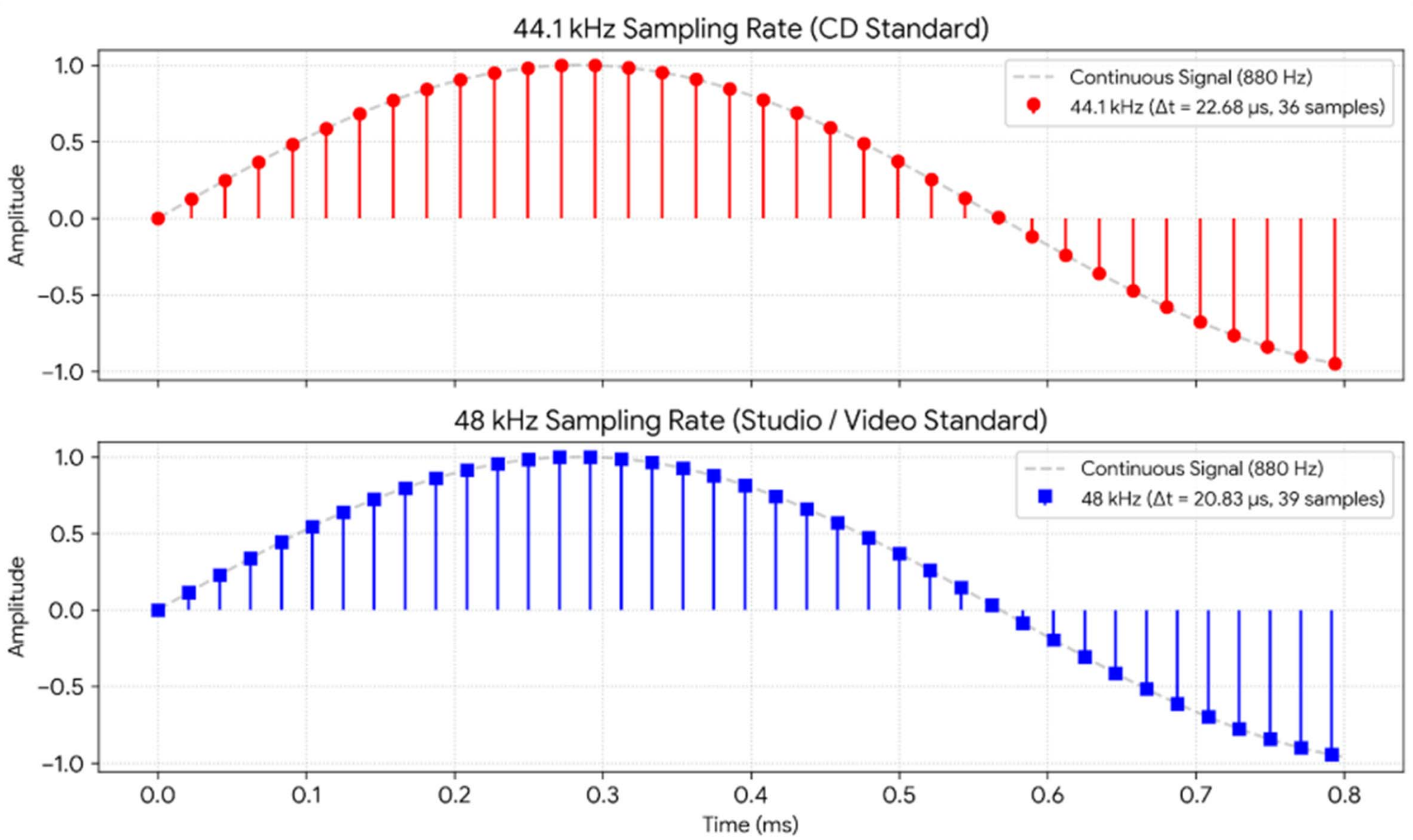

https://music-classification.github.io/tutorial/part2_basics/input-representations.html

# Sampling Rate

- Usually, a model considered signals with a **certain and fixed** sampling rate
  - Data with different sampling rates are not on the same time basis

```
data, rate = sf.read("test.wav") # load audio
```

  - Make sure your data are under the same sampling rate
  - Double check the sampling rate assumed by the models you sourced from GitHub
    - Speech: 16k Hz
    - Music: 22k, 44k, or 48k Hz
  - If we want to ML/DL model trained on signals with one sampling rate to process a signal under a different sampling rate, "**resampling**" of that signal is needed
    - Upsampling
    - Anti-aliasing (low-pass filtering)
    - Downsampling


44.1 kHz Sampling Rate (CD Standard)
Continuous Signal (880 Hz)
44.1 kHz (Δt = 22.68 µs, 36 samples)
Amplitude
1.0
0.5
0.0
−0.5
−1.0
48 kHz Sampling Rate (Studio / Video Standard)
Continuous Signal (880 Hz)
48 kHz (Δt = 20.83 µs, 39 samples)
Amplitude
1.0
0.5
0.0
−0.5
−1.0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Time (ms)

# Resampling

https://docs.pytorch.org/audio/stable/tutorials/audio_resampling_tutorial.html

resampler = **torchaudio.transforms.Resample**( orig_freq=orig_freq, new_freq=new_freq )

resampled_waveform = resampler(waveform)

| Library | Time on CPU (ms) |
|---|---|
| soxr (HQ) | 7.2 |
| scipy.signal.resample | 13.4 |
| soxr (VHQ) | 15.8 |
| torchaudio | 19.2 |

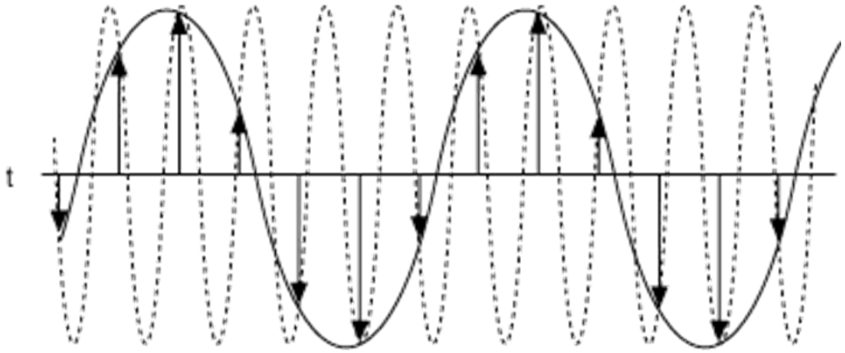
# Sample Rate independent Models

- Usually, a model considered signals with a certain and fixed sampling rate
- Building a **sampling-rate agnostic** model that can take signals sampled at arbitrary sampling rates is an advanced topic

Ref 1: Saito et al, “Sampling-frequency-independent audio source separation using convolution layer based on impulse invariant method,” EUSIPCO 2021
Ref 2: Carson et al, “Sample rate independent recurrent neural networks for audio effects processing,” DAFx 2024

# Nyquist–Shannon Sampling Theorem

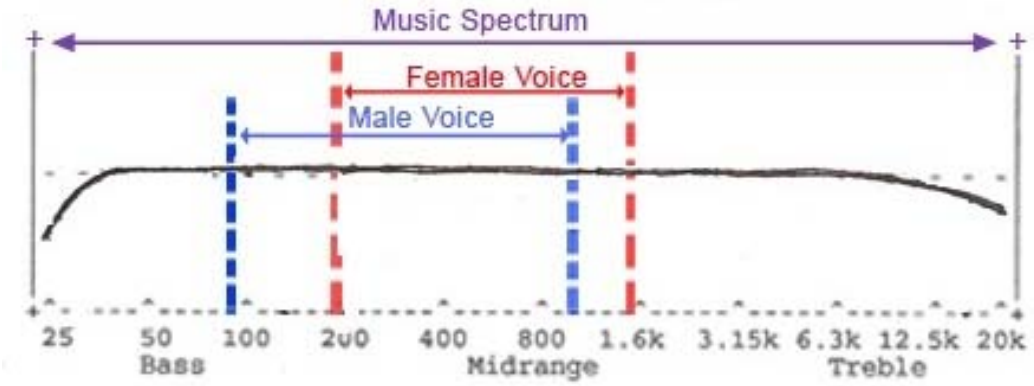
- A signal must be sampled at least **twice** as fast as the bandwidth of the signal to accurately reconstruct the waveform; otherwise, the high-frequency content will alias at a frequency inside the spectrum of interest
- **Sampling freq (Fs) > 2* the highest freq in the signal**




http://zone.ni.com/reference/en-XX/help/370524T-01/siggenhelp/fund_nyquist_and_shannon_theorems/

# Nyquist–Shannon Sampling Theorem

- Telephone audio: 8k Hz
  - Via phone, we cannot hear frequency higher than 4k Hz




https://www.quora.com/How-do-HRT-sex-reassignment-and-other-such-procedures-affect-vocal-production-particularly-the-singing-voice

- **Question**: With Fs=128 Hz, we assume that we don't need to care about frequency higher than __ Hz in brain waves

# Nyquist–Shannon Sampling Theorem

Brainwaves, Frequencies and Functions

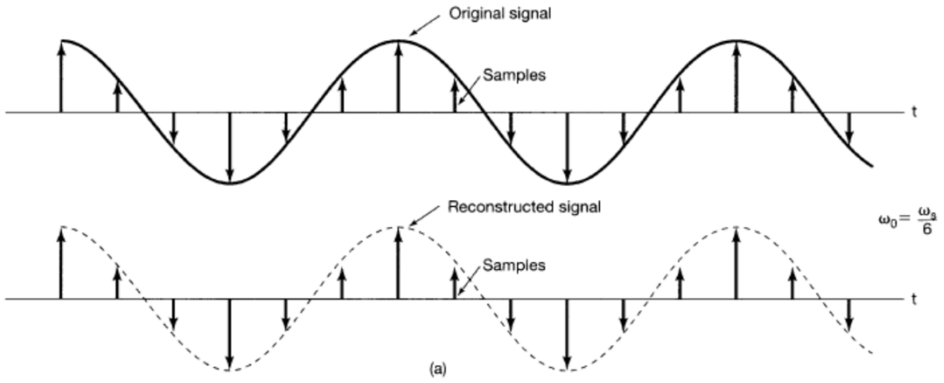
| Unconscious | | Conscious | | |
|---|---|---|---|---|
| **Delta** | **Theta** | **Alpha** | **Beta** | **Gamma** |
| 0,5 – 4 Hz | 4 – 8 Hz | 8 – 13 Hz | 13 – 30 Hz | 30-42 Hz |
| **Instinct** | **Emotion** | **Consciousness** | **Thought** | **Will** |
| Survival<br>Deep sleep<br>Coma | Drives<br>Feelings<br>Trance<br>Dreams | Awareness of the body<br>Integration of feelings | Perception<br>Concentration<br>Mental activity | Extreme focus<br>Energy<br>Ecstasy |

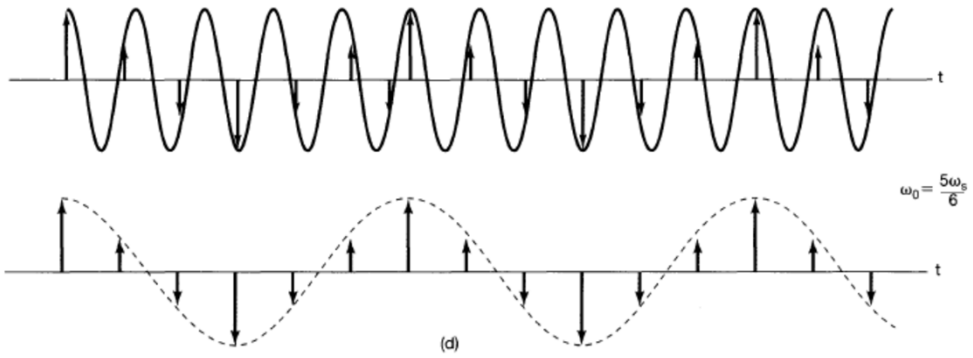
http://altered-states.net/barry/update236/

- **Question**: With Fs=128 Hz, we assume that we don't need to care about frequency higher than 64 Hz in brain waves

# Aliasing

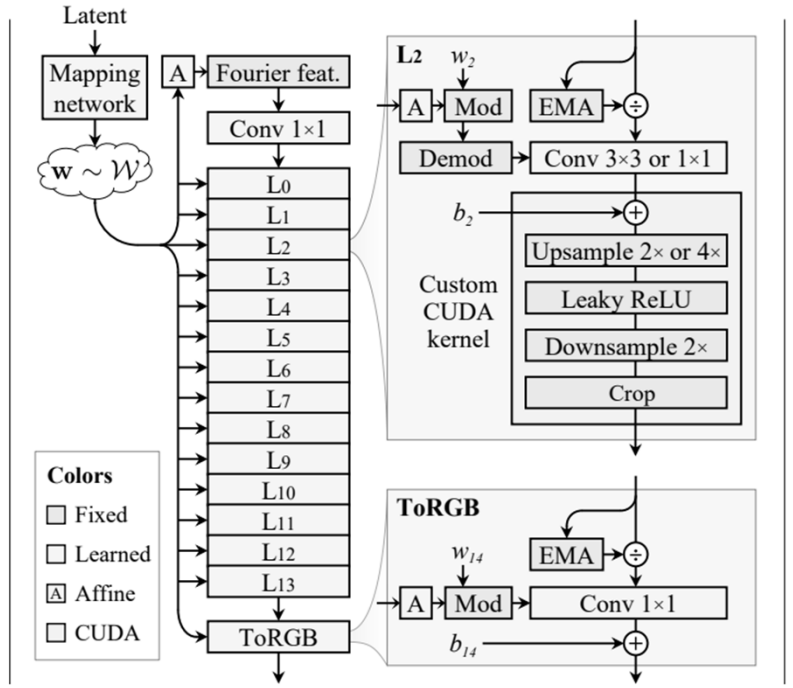
- *High-frequency* information has already **disguised itself** as ("aliased into") a *low-frequency* signal at discrete sample points
- The values of high-frequency and low-frequency components become completely identical at discrete sampling points



# Anti-Aliasing

- A neural network may perform lots of **downsampling** (e.g., pooling)
- Special care is needed for anti-aliasing, e.g., via low-pass filtering (**LPF**)
  - So that high-frequency information is "**removed**" rather than aliased into the low-frequency part



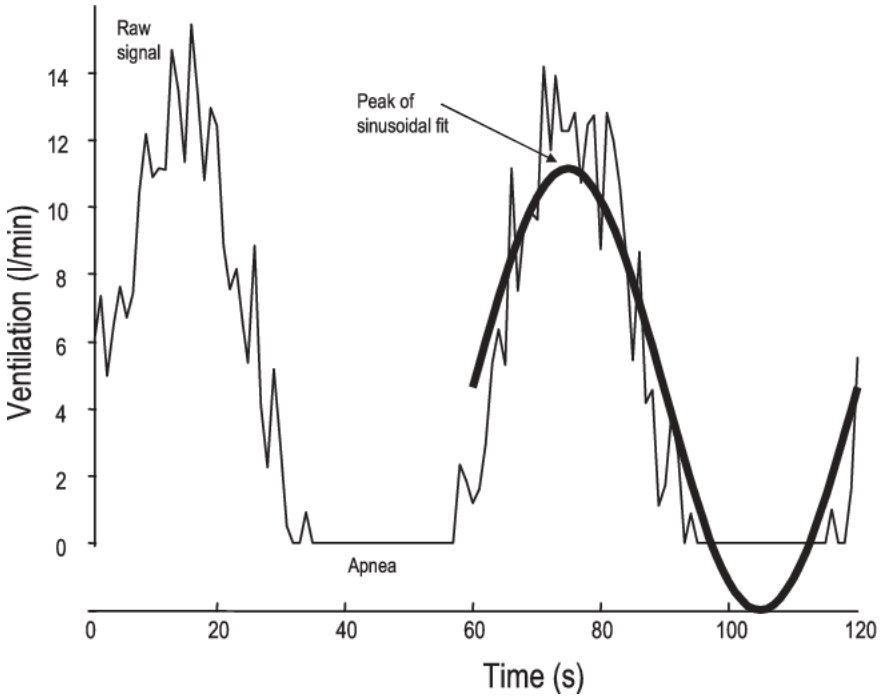

(b) Our alias-free StyleGAN3 generator architecture

https://nvlabs.github.io/stylegan3/

Ref 2: Karras et al, "Alias-free generative adversarial networks," NeurIPS 2021

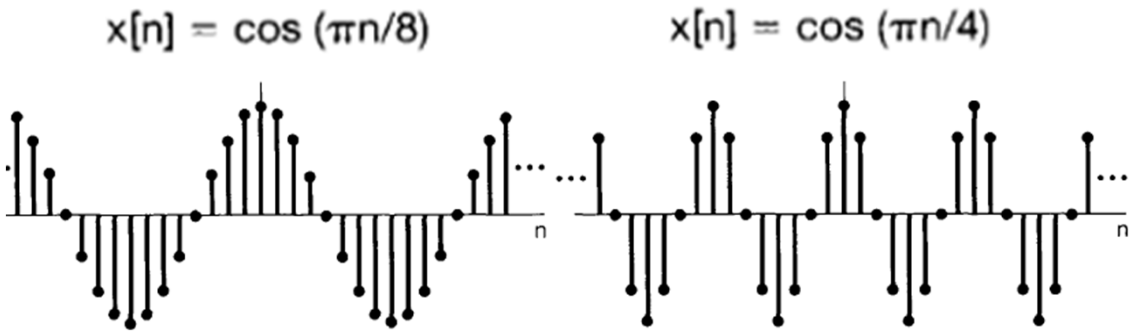
# DTFS: Fourier Series Representation for Discrete-time Periodic Signals

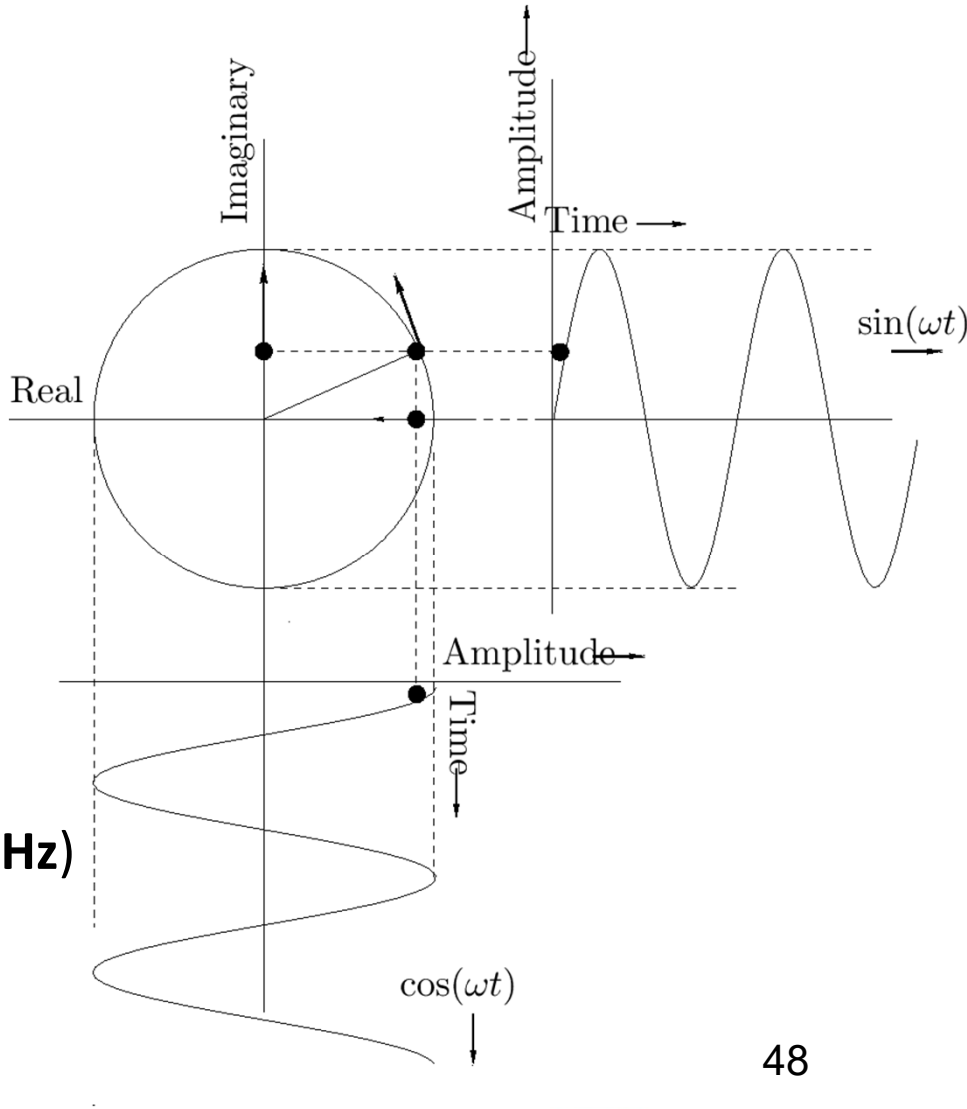
- A discrete-time (DT) **periodic** signal $x[n]$ with period $\boldsymbol{N}$ can be approximated by a linear combination of a harmonically-related family of basis functions $\phi_k[n]$
  - $\phi_k[n] = e^{jk\frac{2\pi}{N}n}$
  - $x[n]$ and $\phi_k[n]$ are all periodic with the common period $\boldsymbol{N}$
- The coefficients of the linear combination, $\boldsymbol{a_k}$ (a.k.a, the *Fourier series coefficients*; DTFS), represents the **inner product** of $x[n]$ and $\phi_k[n]$



# The Basis Functions in DTFS

- $e^{j\omega n} = \cos(\omega n) + j\sin(\omega n)$




- angular or ordinary frequency
  - $\omega$: **angular frequency** (radians per second)
  - $f$: **ordinary frequency** (circles per second; in **Hz**)
  - $e^{j2\pi fn} = \cos(2\pi fn) + j\sin(2\pi fn)$

# DTFS: Fourier Series Representation for Discrete-time Periodic Signals

- For **periodic** signals with fundamental **period $N$**, i.e., $x[n] = x[n+N]$
- The Fourier series coefficients $a_k$ can be interpreted as the **inner product** of the signal $x[n]$ and a basis function $e^{jk\frac{2\pi}{N}n}$, which is complex-valued
- $a_k$ are **complex-valued**
- $a_k$ are also **periodic with period $N$**, i.e., $a_k = a_{k+N}$

**Synthesis** equation: $x[n] = \sum_{k=\langle N \rangle} a_k e^{jk\frac{2\pi}{N}n}$

**Analysis** equation: $a_k = \frac{1}{N}\sum_{n=\langle N \rangle} x[n] e^{-jk\frac{2\pi}{N}n}$

In discrete-time, $e^{jk\omega n} = e^{j(k+N)\omega n}$; so there are **only $N$ unique basis vectors**

**Sum over a set of $N$ successive integers**

# **Discrete-time Fourier Transform (DFT):** Extending DTFS to Deal with Aperiodic, Finite-Duration DT Signals

- For **periodic DT** signal $x[n] \overset{\mathcal{FS}}{\leftrightarrow} a_k$

**Synthesis**: $x[n] = \sum_{k=\langle N \rangle} a_k e^{jk\omega n}$

**Analysis**: $a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk\omega n}$

- For **aperiodic, finite-duration DT** signal of length $L$, $x[n] \overset{\mathcal{DFT}}{\longleftrightarrow} X_k$

**Synthesis**: $x[n] = \sum_{k=0}^{L-1} X_k e^{jk\omega n}$

**Analysis**: $X_k = \frac{1}{L} \sum_{n=0}^{L-1} x[n] e^{-jk\omega n}$

NOTE: While the length of $x[n]$ is $L$, the length of $X_k$ is also $L$

# STFT: Short-time Fourier Transform

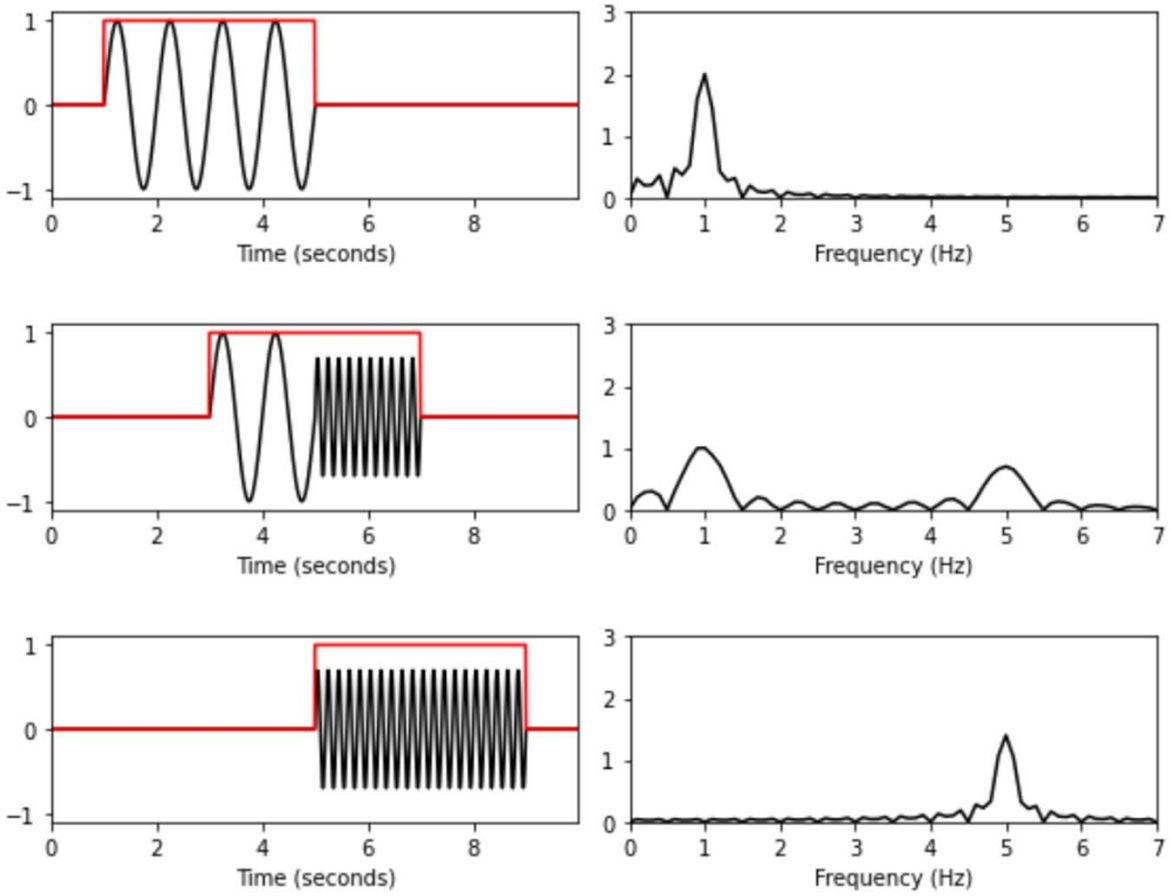
https://www.audiolabs-erlangen.de/resources/MIR/FMP/C2/C2_STFT-Basic.html

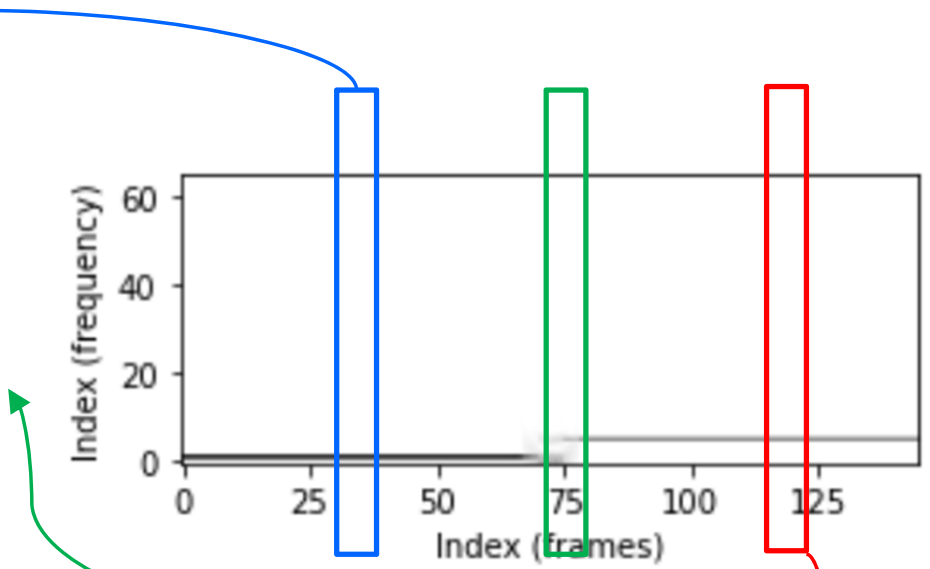
$$\mathcal{X}(m,k) := \sum_{n=0}^{N-1} x(n+mH)w(n)\exp(-2\pi ikn/N)$$

- $x[n]$: real-valued discrete-time signal of length *L* obtained by equidistant sampling with respect to a fixed sampling rate
- $w[n]$: window function of length $N_{\text{win}} < L$
- $H$: "hop size"
- **STFT**: a sequence of **localized DFTs**

# Localized/Windowed FFT

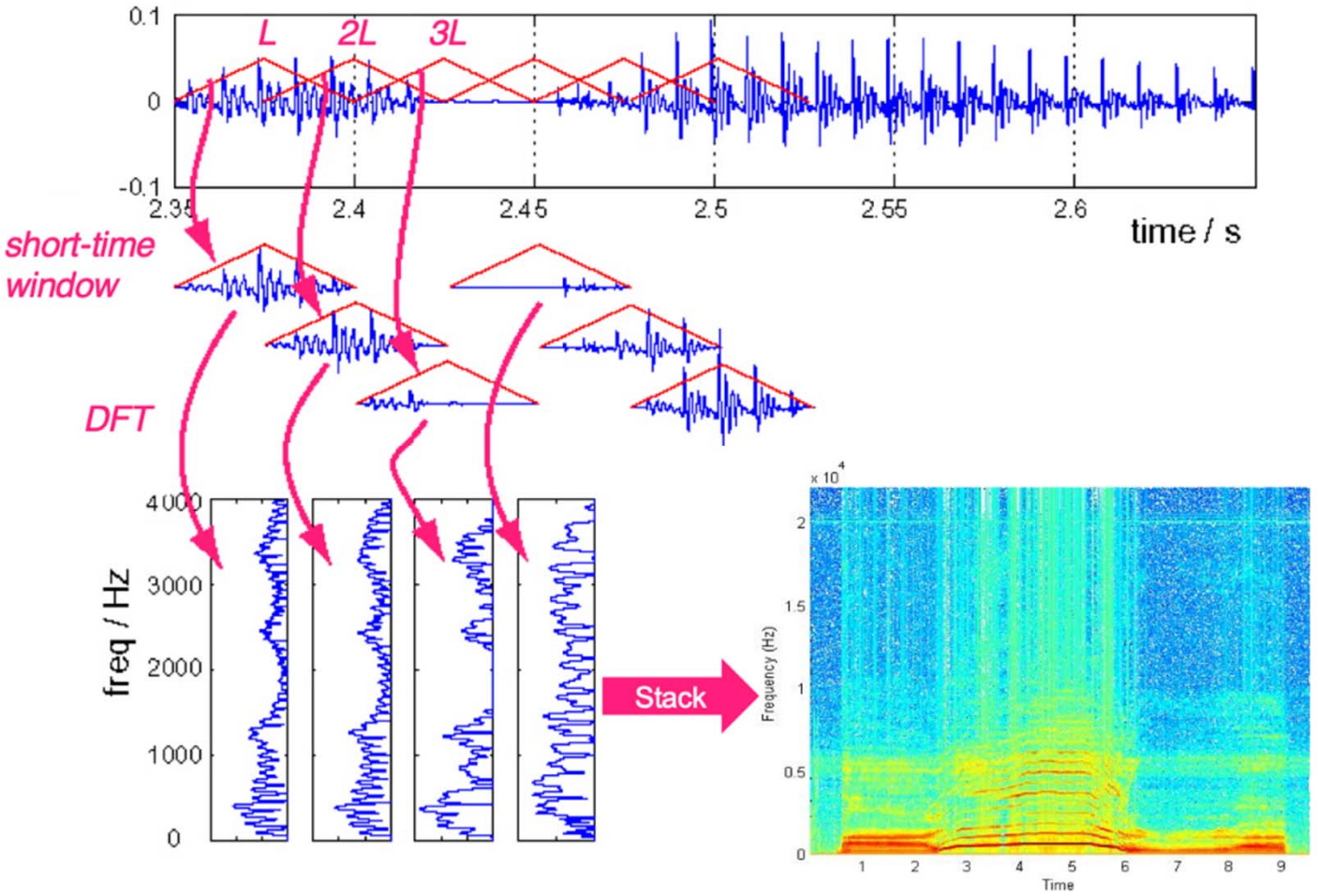
https://www.audiolabs-erlangen.de/resources/MIR/FMP/C2/C2_STFT-Basic.html



# Localized/Windowed FFT

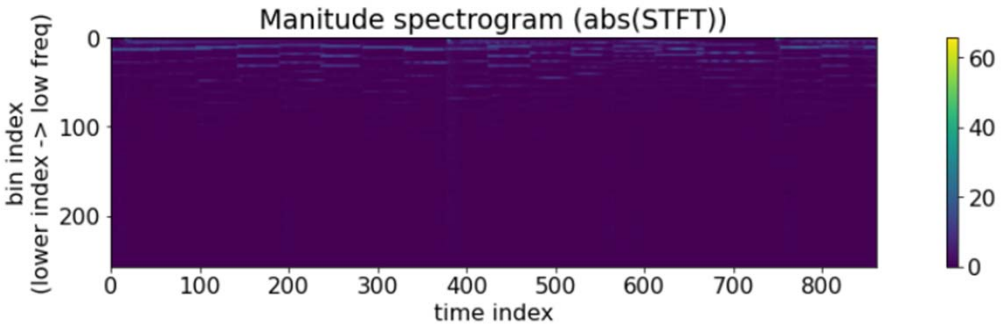
https://source-separation.github.io/tutorial/basics/representations.html

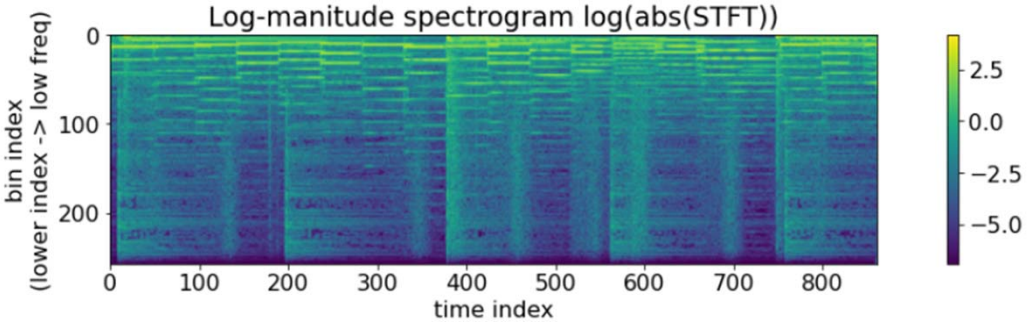


# Magnitude Spectrogram

https://music-classification.github.io/tutorial/part2_basics/input-representations.html?highlight=phase

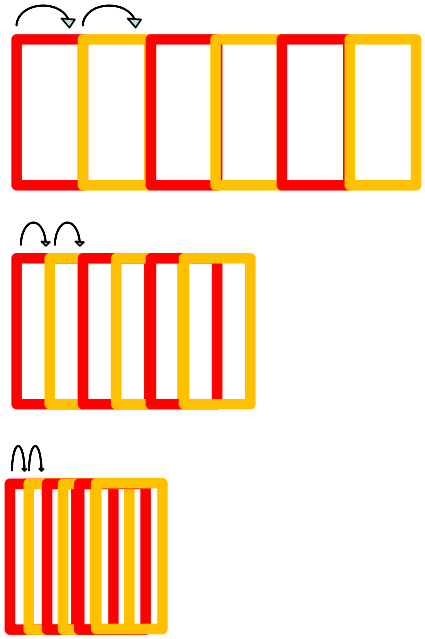
- **STFT**: complex-valued, linear frequency scale
- **Spectrogram**: a time-frequency representation, usually computed via STFT
- **Magnitude spectrogram**: abs(STFT); real-valued, non-negative
- **Log-magnitude spectrogram**: log(abs(STFT)); real-valued, non-negative
- The **phase** information of STFT is often considered less important



# Hop Size in STFT

- Hop size
  - Can be proportional to the window size
    - hop_size = win_size
    - **hop_size = 0.5*win_size**
    - hop_size = 0.1*win_size
  - Can also be not proportional to the window size



# torchaudio. transforms.Spectrogram

https://docs.pytorch.org/audio/main/generated/torchaudio.transforms.Spectrogram.html

```
waveform, sample_rate = torchaudio.load("test.wav", normalize=True)

transform = torchaudio.transforms.Spectrogram(n_fft=800)

spectrogram = transform(waveform)
```

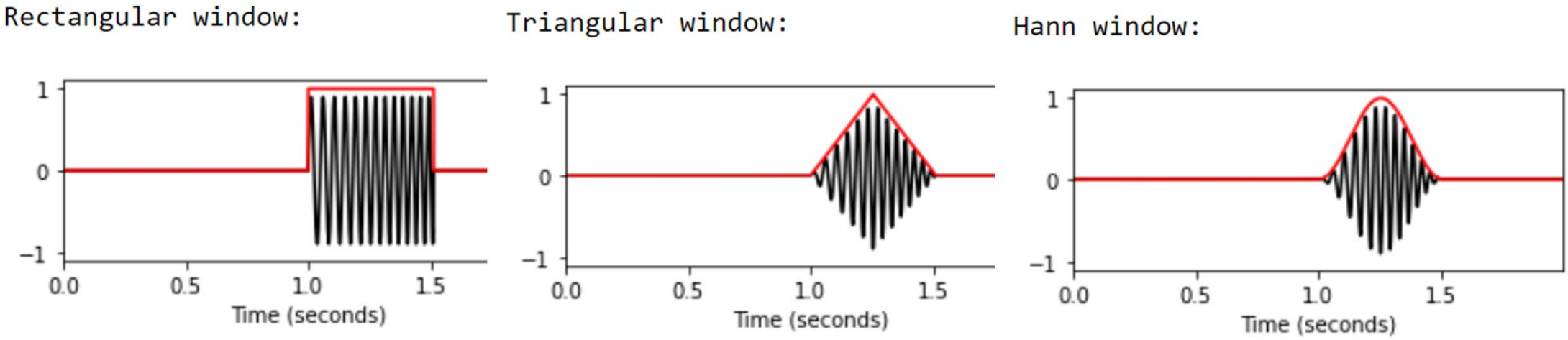
**Parameters**

- **n_fft** (*int*, *optional*) – Size of FFT, creates `n_fft // 2 + 1` bins. (Default: `400`)
- **win_length** (*int or None*, *optional*) – Window size. (Default: `n_fft`)
- **hop_length** (*int or None*, *optional*) – Length of hop between STFT windows. (Default: `win_length // 2`)

# Window Functions in STFT

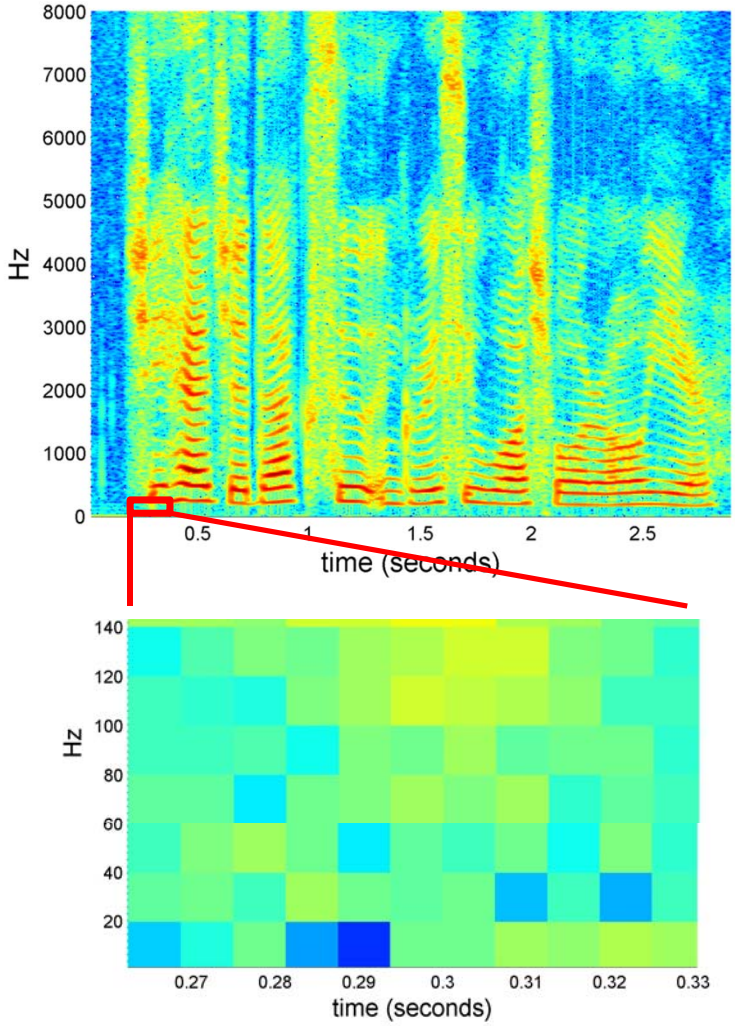
https://www.audiolabs-erlangen.de/resources/MIR/FMP/C2/C2_STFT-Window.html

- Hann window is often used



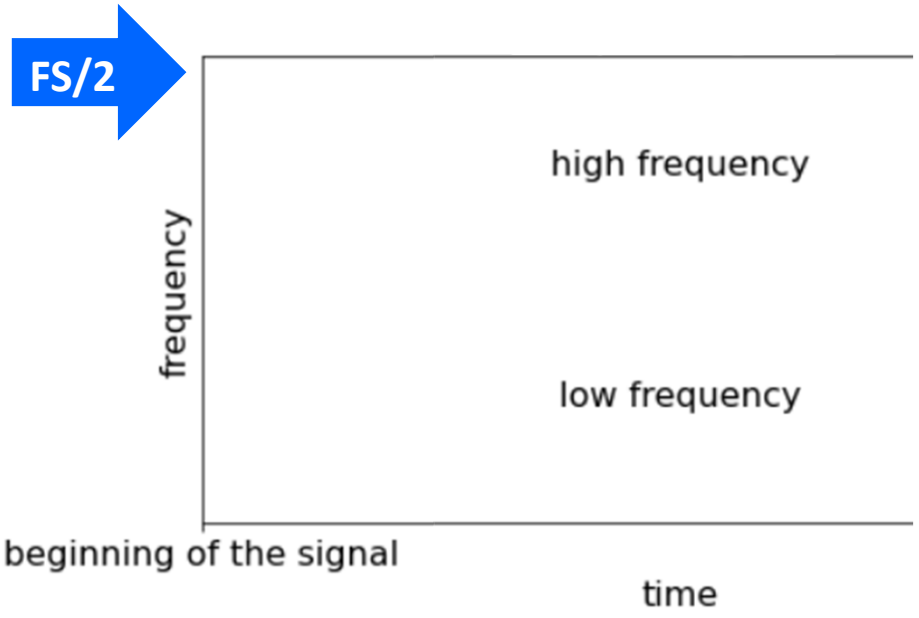
# Frame, Frequency Bin, and Time-Frequency Point

- The magnitude spectrogram can be represented as a **matrix**
- **Frame**: along time
- **Frequency bin**: along frequency
- **Time-frequency point**

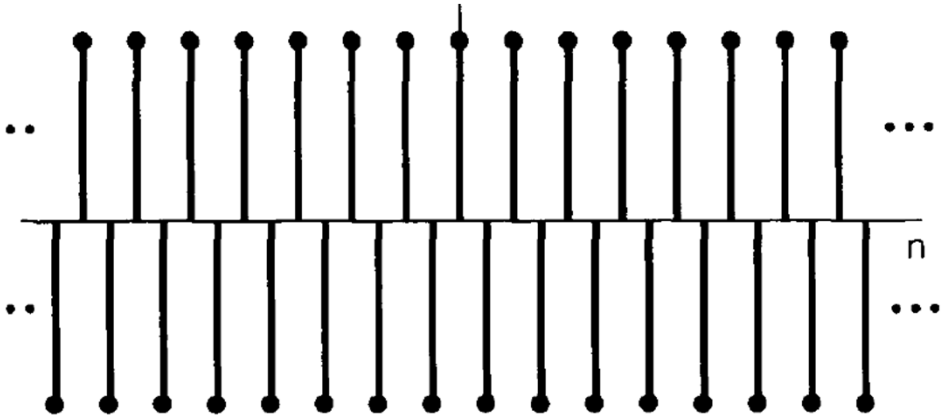


# Nyquist–Shannon Sampling Theorem

- Sampling freq (Fs) > 2* the highest freq in the signal
- **Highest freq bin in STFT = Fs/2**



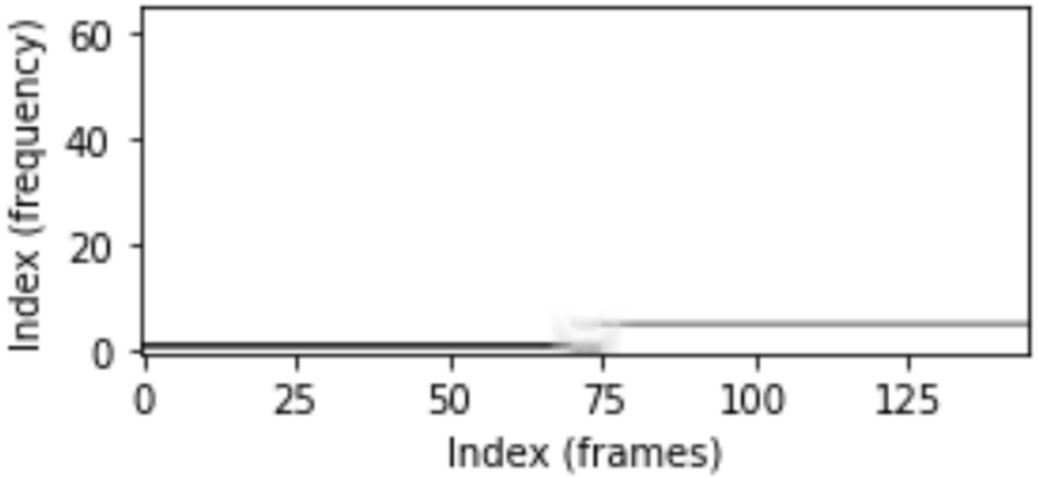

Sinusoid with the highest freq given a fixed FS



# Interpretation of Time and Frequency Indices in STFT

https://www.audiolabs-erlangen.de/resources/MIR/FMP/C2/C2_STFT-Basic.html

$$F_{\text{coef}}(k) := \frac{k \cdot F_{\text{s}}}{N}$$




- **Frequency spacing** (Δfreq): $\frac{F_s}{N}$
- **Temporal spacing** (Δtime): $\frac{H}{F_s}$

$$T_{\text{coef}}(m) := \frac{m \cdot H}{F_{\text{s}}}$$

# Frequency spacing ($\Delta$freq) $= \frac{F_s}{N}$

- $a_k = \frac{1}{N}\sum_{n=\langle N\rangle} x[n]e^{-jk\frac{2\pi}{N}n}$
  - $a_k = a_{k+N}$ for every signal
  - $a_k = a_{-k}^* = a_{N-k}^*$ for real signals due to conjugate property (thus, $|a_k| = |a_{N-k}|$)
- For real signals, we only need to consider $k = 0, \ldots \frac{N}{2}$
  - $a_0 = \frac{1}{N}\sum_{n=\langle N\rangle} x[n]$
  - $a_1 = \frac{1}{N}\sum_n x[n]e^{-j\frac{2\pi}{N}n}$ (the period of the basis function is $N$; lowest frequency)
  - $a_{\frac{N}{2}} = \frac{1}{N}\sum_n x[n]e^{-j\pi n} = \frac{1}{N}\sum_n x[n](-1)^n$ (the period is 2; highest frequency)
  - In total, $\frac{N}{2} + 1$ points for positive frequency

# Size of the Magnitude Spectrogram

- Number of frequency bins (i.e., number of rows in the matrix): $\left\lfloor \frac{N}{2} \right\rfloor + 1$
  - If $N$ is even, this is $\frac{N}{2} + 1$
  - If $N$ is odd, this is $\frac{N+1}{2}$
  - Only need to consider up to $\left\lfloor \frac{N}{2} \right\rfloor + 1$ because the magnitude of DFT output is symmetric for real-valued input signals
- Number of time frames (columns in the matrix): $\left\lfloor \frac{L-N}{H} \right\rfloor + 1$
  - $L$ is the total number of samples in the signal
  - The $+1$ accounts for the first window starting at the beginning of the signal (i.e., initial frame at offset 0)

# torchaudio. transforms.Spectrogram

https://docs.pytorch.org/audio/main/generated/torchaudio.transforms.Spectrogram.html

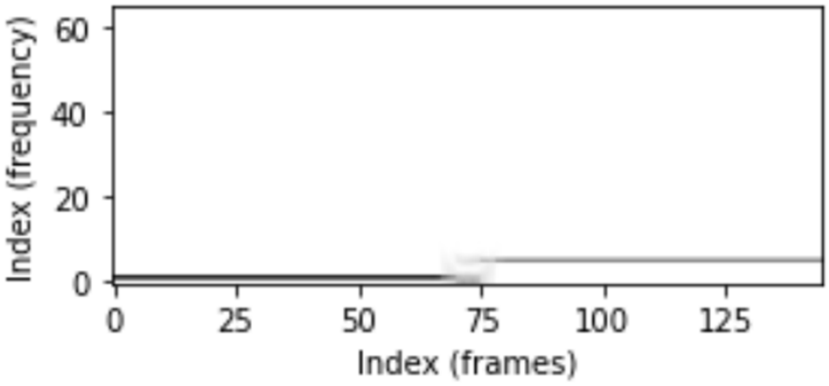
```
waveform, sample_rate = torchaudio.load("test.wav", normalize=True)

transform = torchaudio.transforms.Spectrogram(n_fft=800)

spectrogram = transform(waveform)
```

**Parameters**

- **n_fft** (*int*, *optional*) – Size of FFT, creates `n_fft // 2 + 1` bins. (Default: `400`)
- **win_length** (*int or None*, *optional*) – Window size. (Default: `n_fft`)
- **hop_length** (*int or None*, *optional*) – Length of hop between STFT windows. (Default: `win_length // 2`)

# How about Frequencies in Between?

$$F_{\text{coef}}(k) := \frac{k \cdot F_{\text{s}}}{N}$$



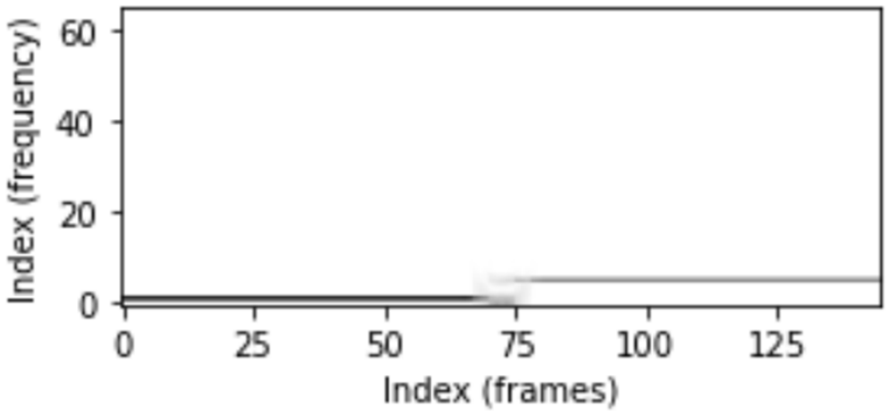

- First frequency bin ($k = 0$): DC
- Second bin ($k = 1$): $\frac{F_S}{N}$ Hz
- An **arbitrary frequency** $f$ rarely falls perfectly on a DFT grid point ($f \neq k\frac{F_S}{N}$)
  - DFT measures how much the input correlates with exact integer harmonic sinusoids
  - the energy for the component at $f$ **spreads across all DFT bins** ("spectral leakage" due to the time-domain windowing operation ["frequency-domain conv with a sinc"])
  - the bin **closest** to the true frequency, $k = \text{round}\left(\frac{f \cdot N}{F_S}\right)$, contains the maximum energy peak, though it does not capture the full signal alone

# How about Frequencies in Between?

- Example

  If $F_s = 44100$ Hz and $N = 1024$

  - First bin: 0 Hz
  - Second bin: 43.066 Hz
  - Third bin: 86.132 Hz




  - 20Hz → peak bin is $k = 0$
  - 30Hz → peak bin is $k = 1$
  - while bin $k$ holds the maximum energy peak, energy still leaks into adjacent bins due to windowing (spectral leakage)

  - **DFT bins are point samples, NOT a frequency range or "intervals"**

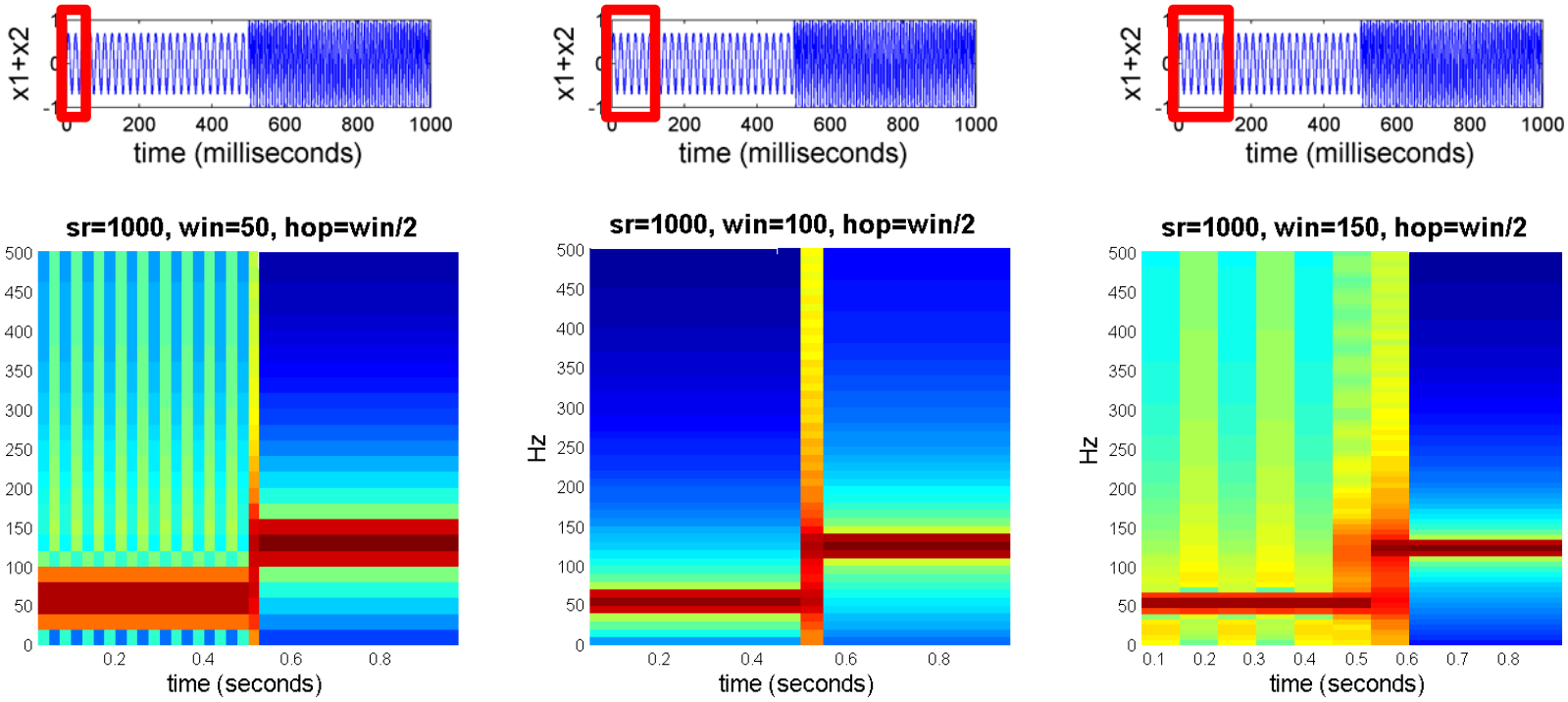
# Trade-off between Frequency/Temporal Resolution in STFT

- **Frequency resolution**: $\frac{F_S}{N}$
  - Longer window size ($N$) → finer frequency resolution (but larger resulting STFT)
    → can localize events along the frequency axis
- **Temporal resolution**: $\frac{H}{F_S}$
  - Smaller hop size ($H$) → finer temporal resolution (but larger resulting STFT)
    → can localize events along the time axis

- If $H = N/R$ (e.g., $R = 4$)
  - Temporal resolution: $\frac{N}{RF_S}$ (no good freq/time resolution at the same time)

# Trade-off between Frequency/Temporal Resolution in STFT

- Different window size (win_size = 50, 100, 150)
- hop_size = win_size/2



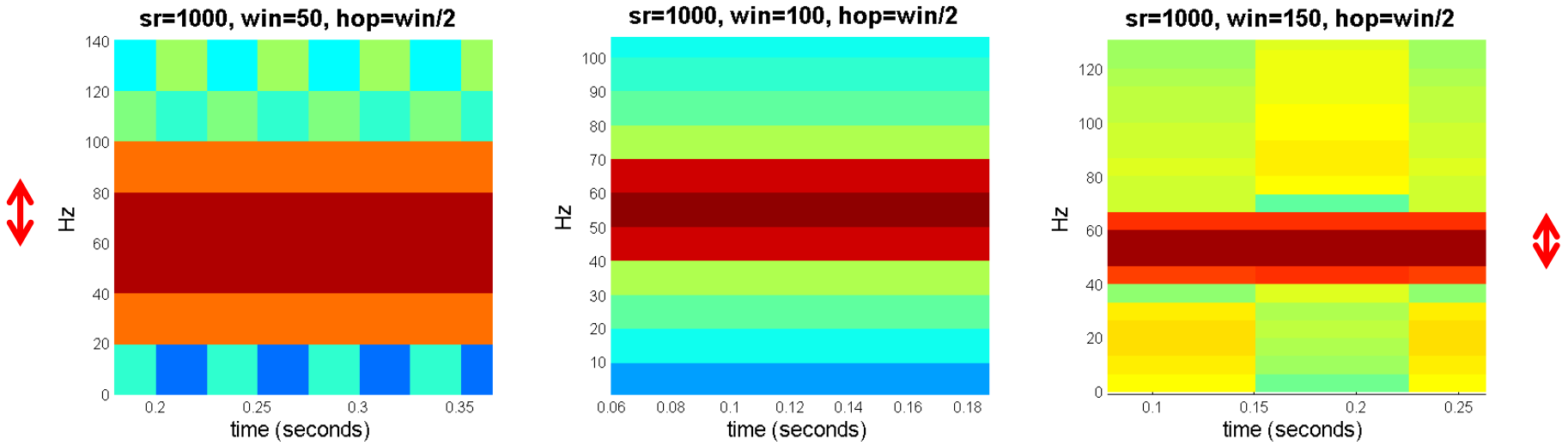

size: 26 x 39

size: 51 x 19

size: 76 x 12

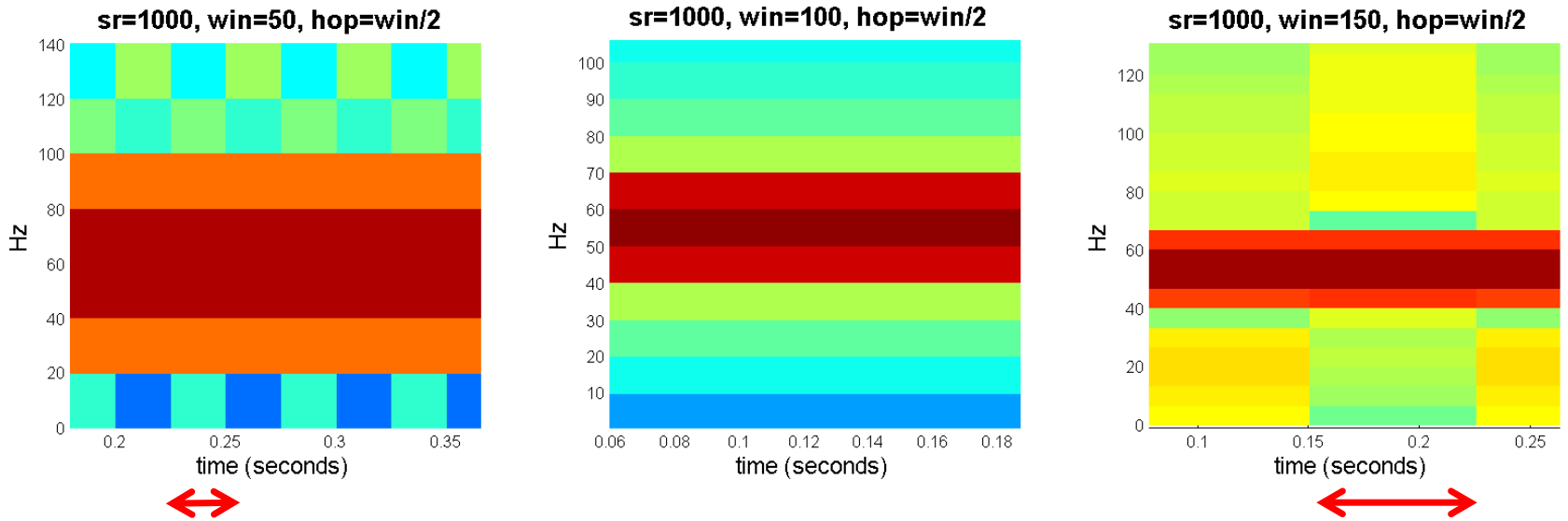
# Frequency Resolution in STFT

- **freq_resolution = Fs/win_size**
  - longer window → better frequency resolution
  - freq_resolution = 20, 10, 6.6667 (Hz), respectively



# Understanding STFT

- **temporal_resolution: hop_size/Fs**
  - longer hop → worse temporal resolution
  - temp_resolution = 25, 50, 75 (ms), respectively



# torchaudio. transforms.Spectrogram

https://docs.pytorch.org/audio/main/generated/torchaudio.transforms.Spectrogram.html

```
waveform, sample_rate = torchaudio.load("test.wav", normalize=True)

transform = torchaudio.transforms.Spectrogram(n_fft=800)

spectrogram = transform(waveform)
```

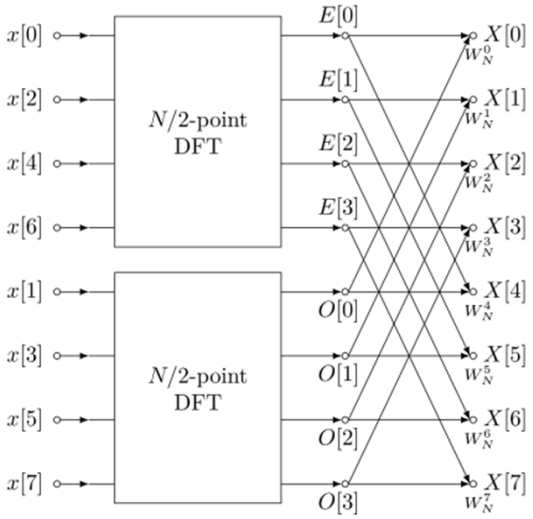
**Parameters**

- **n_fft** (*int*, *optional*) – Size of FFT, creates `n_fft // 2 + 1` bins. (Default: `400`)
- **win_length** (*int or None*, *optional*) – Window size. (Default: `n_fft`)
- **hop_length** (*int or None*, *optional*) – Length of hop between STFT windows. (Default: `win_length // 2`)

# DFT, STFT, FFT (Fast Fourier Transform)

https://www.audiolabs-erlangen.de/resources/MIR/FMP/C2/C2_DFT-FFT.html

- **FFT** is a fast algorithm to compute the DFT; the Cooley-Tukey algorithm requires an array length that is an exact power of 2 (e.g., 512, 1024)
- In practice, we set $N_w = N_{fft}$
- But they can be decoupled
  - $N_w$: physics of the measurement (determines physical resolution)
  - $N_{fft}$: computation and representation (grid density/interpolation)
  - $N_w \leq N_{fft}$ (via zero-padding)



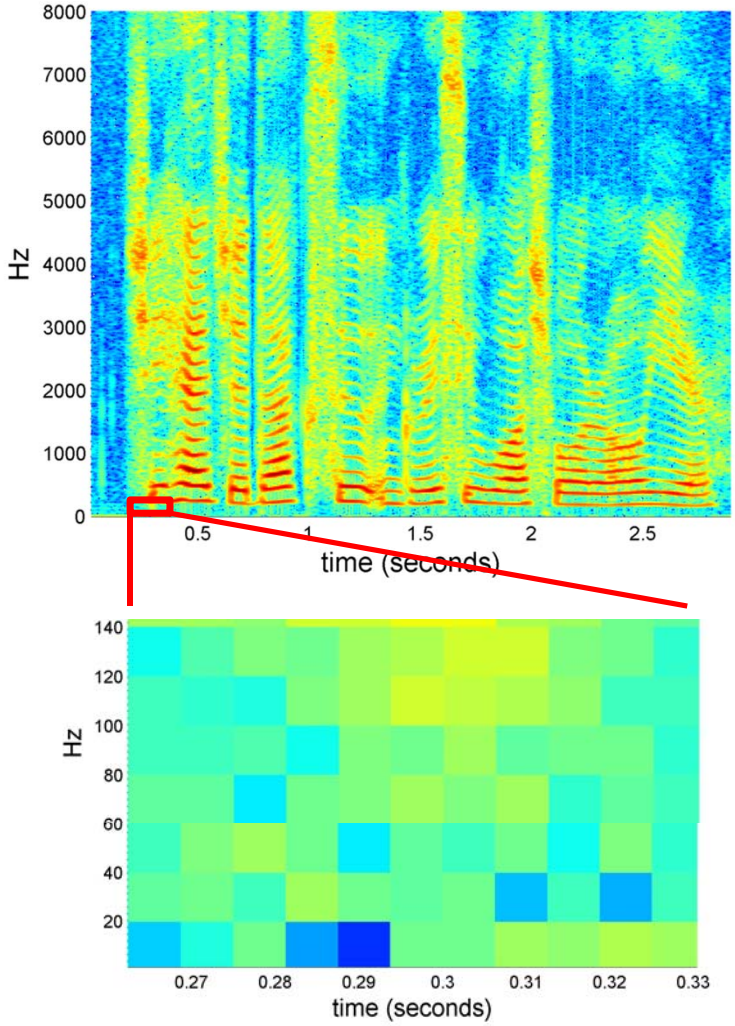

https://en.wikipedia.org/wiki/Fast_Fourier_transform

# Quiz

1. The figure on the top-right is the spectrogram of a signal. What is the underlying sampling rate?

2. The figure on the bottom-right is a zoom-in of the above figure. We can see that the frequency resolution is 20 Hz. What is the window size (in samples)?

3. The temporal resolution is close to 6.6 ms. What's the hop size (in samples), approximately?



# Quiz

4. Given an EEG headset that samples signals at 128 Hz, if we want to be able to discriminate frequency components that differ by 0.5 Hz in frequency, what is the minimal window size (in samples) we need to use? What is the length of such a window in seconds then?
5. Following the previous question, if we further want to discriminate events that differ in time by 0.5 second, what is the maximal hop size (in samples) we need to use?

Brainwaves, Frequencies and Functions

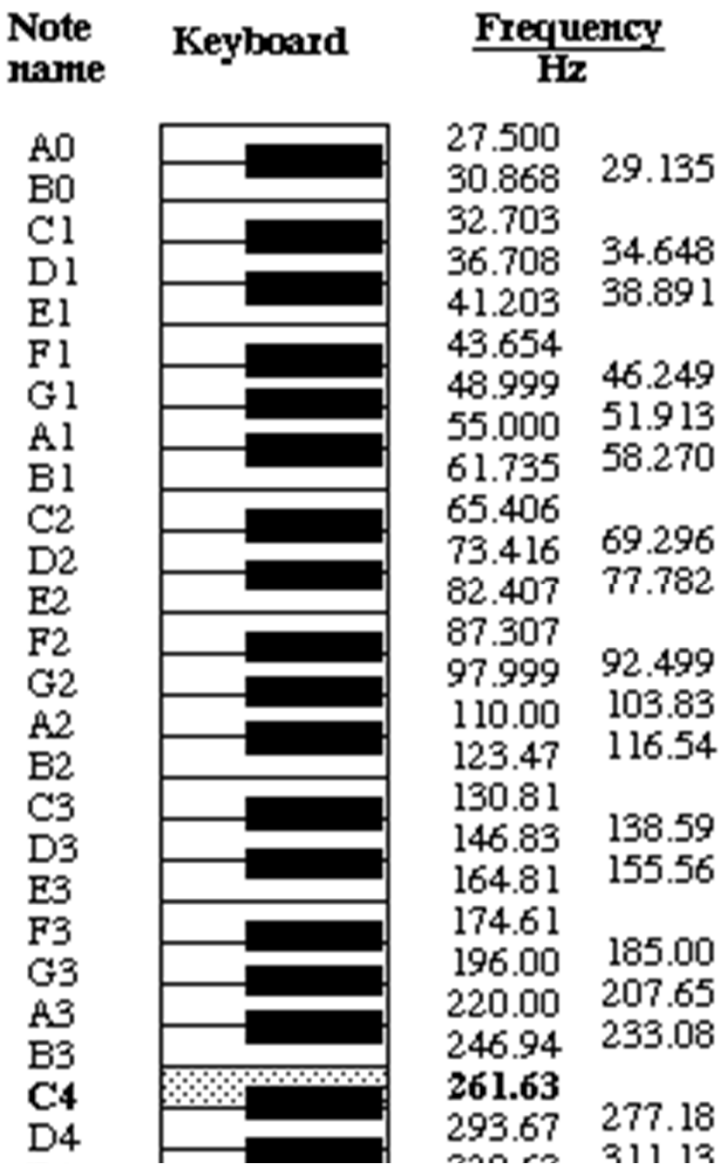
| Unconscious | | Conscious | | |
|---|---|---|---|---|
| **Delta** | **Theta** | **Alpha** | **Beta** | **Gamma** |
| 0,5 – 4 Hz | 4 – 8 Hz | 8 – 13 Hz | 13 – 30 Hz | 30-42 Hz |
| **Instinct** | **Emotion** | **Consciousness** | **Thought** | **Will** |

# Quiz

6. Given a music signal with sr = 44,100 Hz, when we use a window size of 1,024 samples, what would be the frequency resolution?

7. According to the figure on the right, we know that the fundamental frequency (f0) of A1 is 55 Hz. Following the previous question, which bin would be A1's peak bin ?

https://newt.phys.unsw.edu.au/jw/notes.html

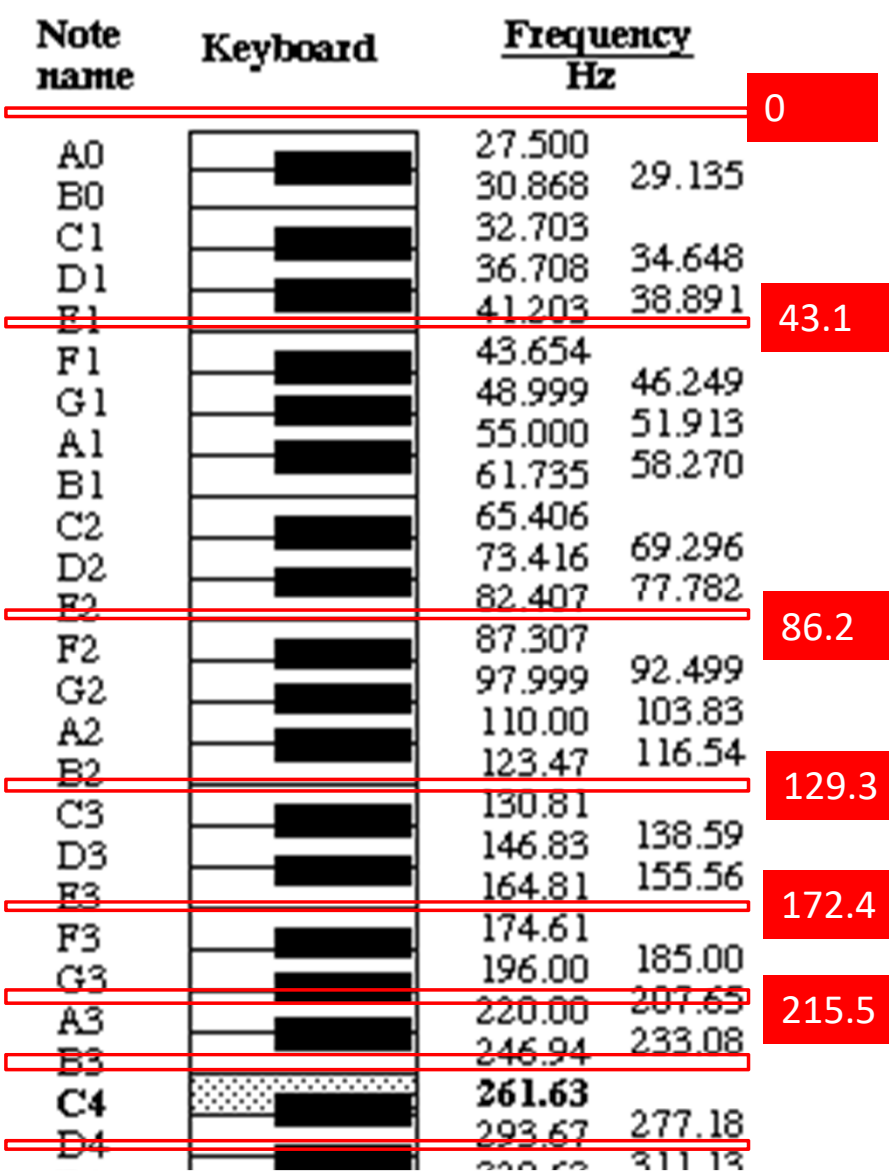


# Quiz

6. Given a music signal with sr = 44,100 Hz, when we use a window size of 1,024 samples, what would be the frequency resolution?
   Sol: 43.1 Hz

7. According to the figure on the right, we know that the fundamental frequency (f0) of A1 is 55 Hz. Following the previous question, which bin would be A1's peak bin ?
   Sol: $k = 1$

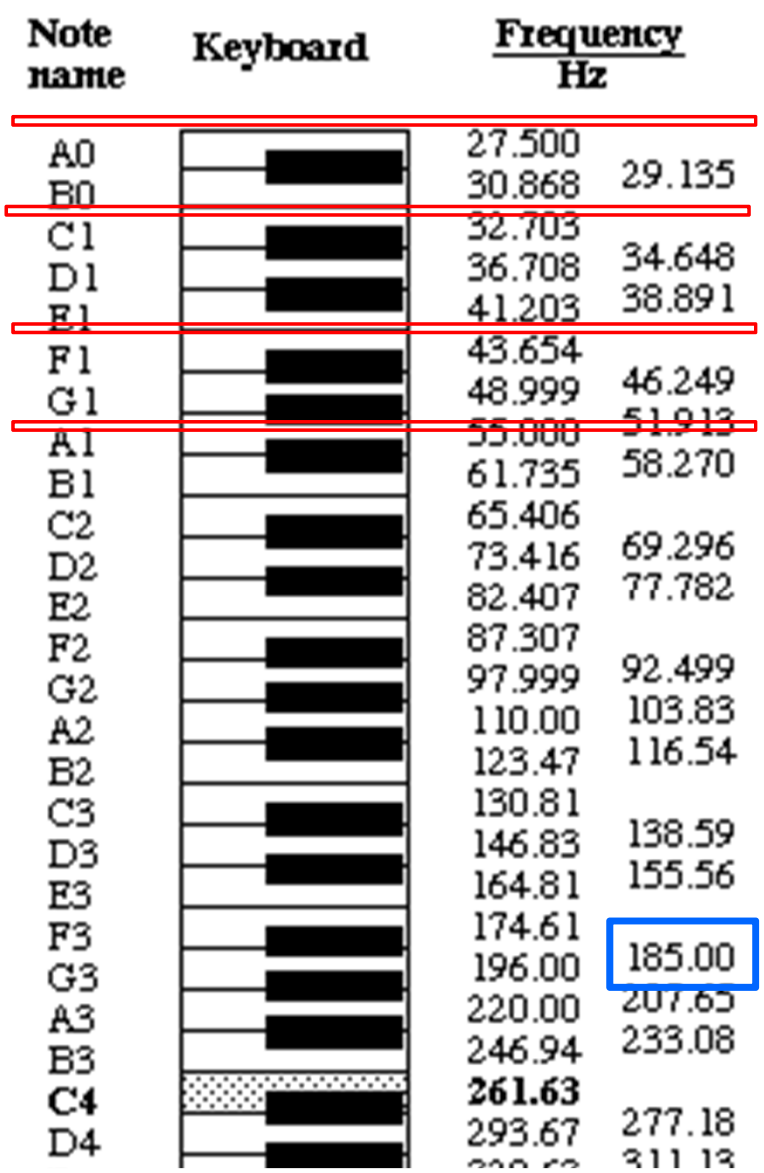
https://newt.phys.unsw.edu.au/jw/notes.html



# Quiz

8. Following the `6` and `7`;
   how if we use a window size of 4,096 samples?

https://newt.phys.unsw.edu.au/jw/notes.html



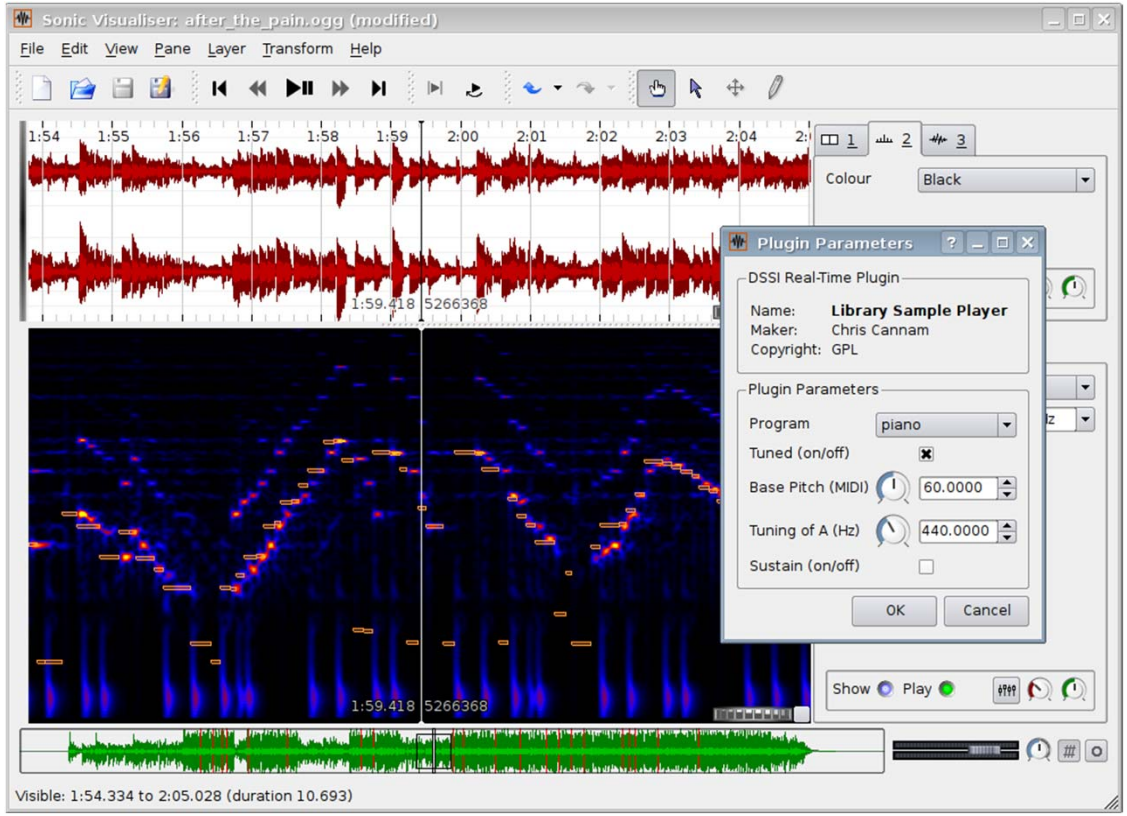
# Summary

- **Sampling rate**, **window size**, **hop size** all matter
  - Frequency resolution
  - Temporal resolution
  - Number of samples, number of frames, and physical time (in milliseconds)

- Make sure the sampling rate is “right”
  - Do “resample” if the sampling rate of your audio files is different from what is assumed by an open-source model
  - Speech: 16k Hz
  - Music: 22k, 44k, or 48k Hz
  - Similarly for window size, hop size, window function, etc

# Appendix: Sonic Visualiser

http://www.sonicvisualiser.org/



# Appendix: Audacity

https://www.audacityteam.org/